AI-ENABLED COCONUT LAND UTILIZATION, WCLWD SEVERITY ASSESSMENT, GERMPLASM IDENTIFICATION AND COMMUNITY INTELLIGENCE ADVISORY SYSTEM

INDIVIDUAL THESIS

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ABSTRACT

Coconut field gene banks and plantations have a limited amount of land to maximize and with conventional visual assessment, there is no standard way to assess how much area is taken up and if there are gaps in the plantation that can support further palms. This research proposes an aerial-image-based land utilization and approximate planting capacity analysis component based on deep learning-based object detection, image processing, and spatial analysis. The public Aerial Images of Palm Trees dataset was converted to the YOLO format, and split into 279 YOLO training images, 42 validation images and 10,398 Palm/Tree bounding boxes. The COCO-pretrained YOLO11n detector was fine-tuned for 50 epochs with 640 × 640 images and a batch size of 8. The best checkpoint achieved 78.2% precision, 82.4% recall, 83.5% mAP@50, and 48.9% mAP@50–95. Tree locations were used to build a circular-buffer occupancy baseline, followed by a dynamic Excess Green (ExG) vegetation mask, elliptical constraints and morphological operations were applied to obtain better estimates of canopy-covered areas. After that, a Euclidean Distance Transform and local-maximum analysis were applied and high-clearance free space regions were identified, and a greedy placement algorithm was used to estimate the additional planting positions under an image space spacing constraint. The model identified 86 Palm/Tree instances in the demonstration aerial image and utilization fell from 32.29% (circular buffer) to 23.81% (with the refined vegetation mask), with 172 candidate free-space regions, and an additional ~55 planting positions identified by a proxy based on the spacing of 148.2-pixels. The results highlight the potential for transforming aerial detections to interpretable land utilization and capacity information. The outputs are subject to interpretation as an estimate in image space to support decisions rather than as an operational planting recommendation, however, due to its current implementation, the model does not include ground sampling distance, local training data for coconuts, agronomic distance calibration, and field validation.

Keywords: Aerial imagery, YOLO11n, land utilization, Excess Green, planting capacity

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I also acknowledge the creators of the publicly available Aerial Images of Palm Trees dataset, which was used for developing and evaluating the aerial Palm/Tree detection model in this component. In addition, I am thankful to Kaggle for providing access to the GPU-enabled notebook environment used for model training and experimentation. Finally, I would like to thank my family, friends, academic staff, and everyone who supported me directly or indirectly during the completion of this research and thesis.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
AP	Average Precision
COCO	Common Objects in Context (image dataset)
CSV	Comma-Separated Values
CUDA	Compute Unified Device Architecture
EDT	Euclidean Distance Transform
ExG	Excess Green index
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
GSD	Ground Sampling Distance
IoU	Intersection over Union
KDE	Kernel Density Estimation
mAP	mean Average Precision
RGB	Red, Green and Blue
RQ	Research Question
SDG	Sustainable Development Goal
TP	True Positive
UAV	Unmanned Aerial Vehicle
WCLWD	Weligama Coconut Leaf Wilt Disease
YOLO	You Only Look Once

1 INTRODUCTION

1.1 Overall project context

Coconut (Cocos nucifera L.) genetic resources are important for the maintenance of varietal diversity and long-term breeding, crop improvement and adaptation to environmental and biological stresses. Thus, there is a need for systematic collection, characterization, evaluation and maintenance of genetically diverse planting materials for the conservation of coconut germplasm. International Coconut Genetic Resources Network (COGENT) has pointed to the need for coordinated international activities on genebank conservation and utilization of coconut genetic resources [1]. The Coconut Research Institute of Sri Lanka (CRISL) has over 100 local and exotic coconut germplasm materials and has conserved germplasm material for use in breeding and evaluation programme to develop improved cultivars with desirable yield and stress-tolerance characteristics [2]. journal.coconutcommunity.org

There are several interrelated information needs associated with the management of a living coconut germplasm collection. Researchers and plantation managers need to identify germplasm within a collection, determine the efficiency of the use of a limited plantation area, track disease and health status of palms, and have reliable information to guide disease management decisions. These activities are frequently performed with individual field observations, manual records and individual analytical activities that make it hard to have an integrated view of plantation and germplasm status. AI-Enabled Coconut Land Utilization, WCLWD Severity Assessment, Germplasm Identification and Community Intelligence Advisory System is a project that tackles this challenge by using four independently developed AI-based tools on a common web platform: germplasm identification, aerial land-utilization analysis, Weligama Coconut Leaf Wilt Disease (WCLWD) severity assessment, and AI-based disease-identification and advisory. Each component meets a different management need and provides information to a shared coconut conservation and plantation-management context.

In the context of this broader research, efficient land use is especially significant, as field gene banks and plantations are limited in space whereas existing palms need to

be retained for an extended period. Space might be needed for recovering missing accessions, adding new germplasm or reorganizing plantation resources. While allotments with clearly open areas can be seen visually, conventional visual inspection does not consistently quantify the canopy occupancy, spatial clearance and potential for further planting. The individual research presented in this thesis, therefore, targets the development of an Aerial Land-Utilization and Approximate Planting-Capacity Analysis component that can combine aerial RGB imagery, object detection by deep-learning, vegetation-pixel analysis and spatial free-space analysis. The proposed method is not only an aerial image to count trees, but it is intended to convert the Palm/Tree locations into useful information such as occupied image sector, available space, high clearance sector and approximate additional planting site. This adds a layer of computation that might be used in addition to plantation management in the field but does not replace the need for real-world measurements and expert judgment for final agronomic decisions.

1.2 Individual component context

The thesis has an individual research part entitled "Aerial Land-Utilization and Approximate Planting-Capacity Analysis". To enhance the use of aerial tree detection not just by providing a count of trees, but to provide useful information for plantation and field gene-bank management. The approach of object detection is well suited to this task since both the object class and its spatial location are extracted from a single image rather than a typical image classification, which would typically only return to an image-level label. The YOLO family is a single-stage detection method that can predict the class of detected objects and bounding boxes directly from an input image [3]. A small YOLO11n model was used in this work, which was fine-tuned using the Ultralytics framework to detect two classes of aerial objects: Palm and Tree [4]. The detector is thus the basis for the following land-utilization analysis, which gives it a spatial basis.

But the area covered by a tree canopy cannot be adequately described by a bounded-box, as background pixels (soil, shadows, gaps between palm fronds, etc.) are also included. The component thus adds further stages of image processing post object detection. A first approximation of the geometry is created by creating a circular

buffer around the centre points of detected trees to represent the canopy geometry. This baseline is then further enhanced by the Excess Green (ExG = 2G-R-B) vegetation index based on the relative dominance of the green colour channel, which helps to differentiate vegetation from other image areas [5]. The developed implementation restricts ExG analysis to the regions of the YOLO detection of tree canopies padded by 100 m/s, and further constrains these areas based on an elliptical mask and morphological operations, ensuring that the vegetation estimation remains linked to the tree canopy, and not to the green areas in the rest of the image. This design yields an easily interpretable approximation of the image area used and available instead of directly using the dimensions of the bounding boxes. The workflow implemented and the four main products (detections, utilisation estimates, free space regions and approximate planting positions) remain the same for both thesis drafts.

The component estimates the occupied vegetation and then performs a spatial structure analysis of the remaining free area. The binary occupied/free space representation is then subjected to a Euclidean Distance Transform (EDT) to find the distance of each available pixel from the nearest occupied vegetation area. Local clearance of a binary image can be effectively represented by distance transforms [6]. Comparatively open areas with potential for further planting can be found by identifying local maxima on the resulting distance surface. These candidate locations are then queued and passed through an iterative greedy placement algorithm which picks up the location with the maximum available clearance, sets the surrounding area as occupied based on a minimum spacing requirement and repeats the process until no new location meets the spacing requirement. Therefore, the component is able to generate four basic outputs: the locations of Palm/Tree instances, the percentage of utilized image area, the percentage of free space (clearance) in the image, ranked into areas with high clearance, and an approximate number of possible additional planting positions and spatial distribution.

Current implementation does not use any other coordinate spaces besides image/pixel. Ground Sampling Distance (GSD), map coordinates, and metric field measurements are not included and the mean detected canopy radius is merely a

temporary image-space spacing proxy. Thus, the planting position proposed should not be taken as an agronomically proven position. Rather, the component is based on the idea of a decision-support technique that shows that aerial object detection can be integrated with the use of transparent image-processing and spatial-analysis techniques, to provide more informative plantation land-utilization information. This restriction is made explicit in both drafts and is key to the interpretation of your component.

1.3 Problem statement

In terms of management, the ability to accurately quantify current tree distribution and determine whether these sites have enough empty space to add more plantation within both plantations and field gene banks is invaluable. While visual interpretation of available spaced between trees can be done when inspecting the field, such a task is incredibly time-consuming, subjective, and difficult to repeat consistently at a large scale. Imagery analysis opens up a much larger scale of the image and has been shown to effectively detect, count, and geolocate individual palm trees from UAV imagery using deep-learning object detectors [7]. Still, object detection alone does not provide enough information for land-utilization planning: a tree count simply reports the number and approximate locations of detected objects but not how much land remains beneath the canopy or if there is suitable space between detected objects.

Moreover, using individual object-detection bounding boxes directly as the occupied area can lead to overestimating land-utilization estimates. A bounding box is a rectangle, which encompasses all canopy pixels, but also contains soil, shadows, and gaps between fronds. As a result, summing bounding-box areas leads to an overestimate of vegetation-area utilization. However, simply calculating total available-area percentage from the sum of all individually calculated object bounding boxes will not suffice, since free space may be narrow gaps between the canopies, and the cumulative gaps between the current canopies are also accounted in that available-area percentage calculation, but those gaps between fronds may not have enough clearance for planting once spacing constraints are respected. Thus,

available-area capacity and plantable-area capacity are different values; this is a key issue in this thesis.

This problem therefore produces a reproducible computing method to turn aerial palm/tree detections into a more operationally meaningful land-use figure by estimating canopy space occupancy, extent of occluded free-space areas, and approximate additional planting positions considering clearance from pre-existing vegetation and previous proposed planting positions. This is complicated by irregular shapes of palm-tree canopies, frond overlaps, colour variations in foliage and illumination, shadows, scale- and density-variations in the fronds, and incomplete information about the actual ground scale from the available images. Accordingly, this work explores whether a hybrid workflow consisting of YOLO11n object detection, vegetation pixel analysis, Euclidean-distance margin- and input-spacing-based clearance estimation, and spacing-aware greedy placement, is able to generate a practicality-centric image-space depiction of plantation utilization and approximate planting capacity. As the current implementation is constructed without regard to GSD or field-calibration for spacing, this capacity figure is as merely an estimate for decision support, and not a field-derived agricultural planting figure.

1.4 Research gap

Previous work has shown that a deep-learning-based object detector can identify, counting, and geolocating palm trees in high-resolution aerial images [7]. While these methods demonstrate that object detection can be a valuable basis for plantation monitoring, detection and counting alone don't directly respond to the land-utilization question raised in this work. The detection bounding box is not a good shape for describing the irregular plant canopy and simply knowing that there is one more tree doesn't inform the amount of land it is occupying. Likewise, knowing how much free image area there it doesn't indicate whether that free space consists of large enough, well-spaced regions to plant more trees.

Furthermore, there is a missing link between the tree-detection outputs and useful spatial decision support. While object-detection tasks like YOLO have proved to be effective for localisation [3], vegetation indices such as Excess Green can offer a

lightweight separation of green vegetation from background [5], and Euclidean Distance Transform methods can be used to measure how much free pixels are left on a grid [6]. All these applications are used as stand-alone processing algorithms, and do not form an integrated pipeline for estimating planted area occupancy and estimated plantings capacity. The literature review for this project provided no directly comparable approach: combining Palm/Tree-detection, vegetation-mask refinement, distance-based free-space analysis and iterative spacing aware placement in an understandable pipeline.
This work contributes to filling that gap by combining these subproblems within a sequential image-analysis pipeline. The individual components are; the initial predictions for actual spatial locations of Palm and Tree instances by YOLO11n detections; use of a vegetation mask based on an ecologically relevant vegetation index to truncate the imprecise initial occupancy estimate; a Euclidean Distance Transform to turn the spatially contiguous free pixel volume into a discontinuous surface of crop clearance; and a greedy algorithm that parameterizes that cropped space into an estimate of the count and spatial positioning of the remaining planting spots. The benefit of this work thus is not in training a novel object detector, or in incorporating a new vegetation index, but in the thoughtful integration of existing methods to turn aerial detections into land-interpretation and planting-capacity estimates.

1.5 Aim and objectives

Aim

Objectives This study aims to design and test an aerial-image-based land-utilization and estimated-planting-capacity analysis method for coconuts plantation and field gene-bank management. Targeted outcomes include 1) identifying existing Palm/Tree instances from aerial images, 2) estimating the percentage of image area covered by vegetation, 3) locating spatially relevant free-space regions, and 4) estimating the number and locations of available planting spots. The proposed method involves using YOLO11n object detection in combination with vegetation-mask-clarification, Euclidean-distance-based clearance computation, and the

spacing-sensitive placing to evolve the text-book aerial tree-detection results into land-utilization decision support data.

Objectives.

1. To develop and evaluate an aerial Palm/Tree detection model using YOLO11n.

This experiment aims to convert the annotated aerial dataset into a YOLO dataset, divide the data into training and validation sets, fine-tune the pretrained YOLO11n model and test the detection performance of the Palm and Tree instance search by precision, recall, mAP@50 and mAP@50-95.

2. To estimate the land area occupied by detected tree canopies from aerial imagery.

The locations of the identified trees are then input to produce an initial estimate of occupancy using a circular geometric buffer. This presents a basic approximation of the space which may be occupied by existing Palm and Tree canopies.

3. To improve land-utilization estimation using image-based vegetation analysis.

An Excess Green vegetation mask, combined with elliptical constraints, followed by a morphological operation is implemented in the detected regions in order to more accurately reflect irregular vegetation canopy shape and to minimize the background region covered by the first geometric approximation.

4. To identify suitable free-space regions based on spatial clearance from existing vegetation.

Local-maximum and Euclidean Distance Transform analysis quantify the distance of available pixels from the regions of occupied canopy, allowing for the identification of relatively open areas that may have sufficient space to accommodate more planting.

5. To estimate approximate additional planting capacity using a spacing-aware placement method.

A greedy placement algorithm is used on the free-space map to incrementally suggest new planting locations and image-space distances to existing plants and to

previously suggested locations. The locations and analysis outputs of this process are collated and prepared for input into a larger coconut-management platform.

1.6 Research questions or hypotheses

These are the key questions that underlie this work, both with regard to the performance of the aerial object-detection model, and the applicability of the later image-processing and spatial-analysis pipelines. They include, for example, the ability to detect Palm and Tree instances reliably from aerial images; the use of vegetation-based refinement to offer a more meaningful measure of occupied land than simple geometries; and the conversion of that free space into an estimate of planting capacity, accounting for spatial clearance.

1. **RQ1:** How accurately can a fine-tuned YOLO11n detector locate Palm and Tree instances in the aerial dataset, as measured by precision, recall and mean Average Precision?
2. **RQ2:** Can pixel-level vegetation analysis inside detected regions refine the coarse occupied-area estimate produced by circular buffers around tree centres?
3. **RQ3:** Can distance-based free-space analysis combined with a greedy spacing rule turn raw available area into a more meaningful count of additional planting positions?
4. **RQ4:** What limitations arise when utilization and capacity are estimated entirely in pixel space, without real-world ground scale?

1.7 Scope limitations and assumptions

This study is scoped to the detection of Palm and Tree instances in RGB aerial imagery and the estimation of image-based land-utilization and estimated planting-capacity. The detector is trained and tested using the open access Aerial Images of Palm Trees data set and the full land-analysis pipeline is demonstrated on a separate high-resolution aerial image also from the Aerial Images of Palm Trees data set. The proposed pipeline encompasses object detection, circular-buffer, occupancy estimation, greenness-based vegetation-mask refinement, Euclidean Distance

Transform analysis, free-space candidate detection, greedy planting-capacity estimation, and exporting of the model, image and coordinate based results with scope for integration into the broader coconut-management platform.

Several limitations are identified by this research. The training data are not coconut-specific nor collected from the Sri Lankan coconut gene banks, and a classifier trained on this data cannot be expected to only classify coconut palms. The developer vegetation mask is empirically derived and has not been validated against manually annotated canopy mask training data. The complete land-use and capacity analysis was only demonstrated on a single, test image so the method's robustness against different plantations, seasons, imaging conditions and structure was not evaluated. The method does not account for soil values, topography, drainage, irrigation, roads, built structures, separation of plots, root-zone intensities, or accession-specific planting densities. Most significantly, all distance and area measurements are made within pixel space as we did not have a Ground Sampling Distance or metric calibration data. Accordingly, the pixel spacing value of 148.2 pixels used by the greedy algorithm is only a provisional approximation based upon the average detected canopy radius value and should not be interpreted as an agronomically calibrated planting distance.

The work therefore makes certain assumptions. These include that the public Palm/Tree dataset can be considered appropriate as an instance of palm canopy images to develop proof-of-methods; green-dominant pixels within detected regions are assumed to be part of vegetation of type within the detected canopy; and the mean of detected canopies width as the image-space spacing proxy in absence of a metric ground scale. This proof-of-method workflow can thus be technically assessed and the appted utilization and planting-capacity estimates released as decision-support products and not as field-ready planting guidance until coconut-appropriate local data, in-situ calibration, and domain expert input is available.

1.8 Individual contribution

This research's individual contribution is the end-to-end development of the Aerial Land-Utilization and Approximate Planting-Capacity Analysis component, from the creation of the training dataset to the model training to the final land analysis from geo-coded aerial images and creating output that can be used by other parts of the system. The portion of this work that I undertook is the first step, which involved processing the public aerial Palm/Tree dataset, including converting the original bounding-box annotations to YOLO format, defining the Palm and Tree classes, and creating a train/validation split at the image level. I configured, fine-tuned, and tested a YOLO11n detector on the selected aerial dataset. This involved monitoring training progression, choosing the optimal checkpoint, assessing the overall and class-specific precision, recall, mAP@50, and mAP@50-95, and inferring on a high-resolution demonstration image.

Apart from object detection, the most significant contribution was the development of the processing procedure to produce land-utilization information from detected locations. The first simple geometric occupancy approach was created by placing circular buffers around detected center points. Visual assessment revealed the occupancy of the buffers containing substantial amounts of background does not canopy. The approach was then extended by using an Excess Green vegetation mask, adding padding around detection regions, constraining the occupancy using elliptical boundaries, and using morphological closing and opening operations. A vegetation image was created using the refined vegetation representation and a Euclidean Distance Transform free-space analysis was implemented including local-maximum detection for free-space ranking by maximum distance from vegetation. Subsequently, a greedy iterative planting-position algorithm was created for this free-space that found new planting locations while preserving a user-specified image-space spacing constraint between lines of vegetation and previously located open-space. This iterative development is an important part of the contribution as it demonstrates progressive improvement of the extracted land management information.

On the individual level, the combined work involved visualization and interpretation of outputs, conversion of candidates and final planting position into CSV files, preservation of the trained detector checkpoint, specification of implementation parameters and preparation of artefacts for upload to the wider coconut-management platform. Of particular importance was the need to ensure transparency of the steps so that the resulting bounding-box object detections, vegetation mask, free-space candidates and final capacity estimate can all be evaluated separately. The individual contribution is therefore not only the trained object detector but the entire analytical pipeline to produce occupied area, available area, free-space-candidates and (estimated) remaining capacity from aerial imagery.

1.9 Thesis structure

The thesis is organized in nine chapters, which sequentially describe the background of research, design process, evaluation and results of the Aerial Land-Utilization and Approximate Planting-Capacity Analysis component. Chapter 1 presents the introduction to the research in general, the component in particular, the problem statement, research gap, aim and objectives, research questions, scope and limitations, assumptions, and individual contribution of the researcher. Chapter 2 discusses literature and technical background relevant to the management of coconut germplasm, aerial Palm/Tree detection, vegetation-index based canopy analysis, and distance-based free-space assessment. Chapter 3 describes the research methodology, which includes the dataset, data preparation, training of YOLO11n, evaluation metrics, vegetation-mask generation, free-space analysis, and planting-capacity estimation. Chapter 4 details the product and solution design, including functional requirements, component architecture, design alternatives, and key technical decisions. Chapter 5 documents the implementation environment, component implementation, integration considerations, verification procedures, and artefacts related to deployment. Chapter 6 highlights the experimental and technical results derived from the detector validation and land-utilization analysis. Chapter 7 discusses the results in relation to the research questions, the existing literature, and the research contribution, the product contribution, limitations, and threats to

validity. Chapter 8 deliberates on the product life cycle activities such as the commercialization, sustainability, scalability, compliance, and contribution to relevant SDGs. Chapter 9 summarizes the findings and concludes the research, assesses the research and individual researcher's objectives, discusses the individual research and product contribution, and details recommendations for future development and validation.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The purpose of the review of literature was to build the theoretical and technical basis for the Aerial Land-Utilization and Approximate Planting-Capacity Analysis component and to justify the choice of implementation methodologies. The scope of the review was much narrower than a general coverage of remote-sensing or agricultural computer-vision literature. The approach of remote sensing and computer vision technology in respect of the research problem, design and processing workflow, was specifically set out with respect to that. The review comprised four broad areas, namely: conservation of coconut germplasm and plantation land management; remote-sensing detection of aerial Palm/Tree with a deep learning methodology; Vegetation detection by colour, morphological image processing; and land-occupation image-based free-space analysis as a spatial-clearance estimation.

Priority was given to journal or conference paper publications, original methodological work, formally available institutional sources and the publication of the aerial palm-tree dataset being used. For coconut genetic-resource management, sources on the COGENT project and published material from the Coconut Research Institute of Sri Lanka were used to inform the agricultural background to the work [1], [2]. For research on detection and enumeration of aerial palm-trees, sources on the application of deep-learning-based object detection to plantations were used [7], as well as the original YOLO literature which underpins the nature of single-stage object detection [3]. Sources on the Excess Green vegetation index were used in support of colour-based canopy-refinement [5], with materials describing Euclidean Distance Transform algorithms used in support of spatial data extraction on the available image area [6]. Software documents were used only where no formal publication existed, most notably the particular instantiation of YOLO11 used for this work [4].

This left the scope of the review limited to methods that can have a direct impact on research design and application decisions. The broader areas of research including

multispectral remote sensing, LiDAR canopy reconstruction, complete GIS platform, soil-suitability modelling and other novel approaches to agricultural optimization were not reviewed in detail as they fell outside of the implemented scope of this component. This approach maintains a strong link between the literature, research gap, methodology and final assessment of the proposed aerial-image-based land-utilization framework.

2.2 Domain and theoretical foundations

The land-utilization component that is suggested here is formed from a suite of concepts from the fields of agricultural image interpretation and computer vision. Based on the necessity of the management of coconut genetic resources, the idea of object detection and scene understanding is supported on the concept of detecting or locating trees, quantitative metrics on measuring the quality of the detector, methods of colour based vegetation analysis for estimating canopies coverage, and distance based spatial analysis for estimating vacant free space. These concepts are the fundamentals of the sequential processing that was carried out in this research.

2.2.1 Coconut genetic resources and land use

Importance of coconut genetic resources for genetic diversity, future crop improvement, breeding, adaptation to changing environmental conditions and ex-situ conservation. The International Coconut Genetic Resources Network has been advocating the collection, conservation, characterization, and use of coconut germplasm [1]. In Sri Lanka, coconut germplasm is conserved in the Coconut Research Institute along with recently introduced germplasm for evaluation and breeding [2]. These were generally maintained as living field genebanks, i.e., for every living accession there is a limited land space in the coconut gardens. Making effective use of this land is thus a practical consideration. Coconut palms are perennials, occupying a planting position for decades, and the development of their crown impacts on the available space around an individual palm. Gaps or underused areas in a plantation or field gene bank may therefore be potential sites for future planting, replacement, or expansion, but requires more information than an inventory of existing palms: the distribution of trees, the approximate occupancy of crown space, and the size of gaps in the existing vegetational cover. This is the main reason

for being able to extract this kind of land-utilization information from aerial imagery quantitatively, rather than rely solely on the visual assessment of a trained observer.

2.2.2 Object detection and the YOLO family

Object Detection is a computer-vision task that predicts objects present and their locations in an image. Whereas an image classification system would typically assign a label to the full image, object detection systems output class predictions for individual instances as well as spatial information about those instances in the form of bounding boxes. This makes it a fitting choice for aerial plantation images as many images contain dozens of trees which need to be located before spatial analysis can begin.

The original YOLO architecture posed object detection as a single regression problem, where bounding boxes and class probabilities are regressed directly from the entire image [3]. This single-stage approach was particularly attractive to efficiently deployed applications. Later generations of YOLO have also evolved their network architectures and training strategies, but with similar end-to-end design as the original, single-path detection network. The agricultural literature has further shown that deep-learning detectors can be applied for counting palms (or other crops) in low-altitude aerial or remotely sensed imagery [7], [8].

This implementation uses YOLO11n, the smallest nano-scale version available in the Ultralytics YOLO11 implementation [4]. Rather than training the detector from random initializations, it is initialized with pretrained weights. The pretrained weights, in this case, were originally trained on the COCO object-detection dataset, a large-scale benchmark dataset with diverse visual object classes [11]. Transfer learning is leveraged here to transfer generic visual object recognition features extracted from a large source dataset towards the more specialized two-class aerial detection problem at hand. In this component, the detector is specialized for class labels Palm and Tree, and the output bounding boxes serve as the spatial input to the subsequent canopy-occupancy and free-space-analysis step.

2.2.3 Detector evaluation metrics

Classification accuracy doesn't capture the quality of object-detection performance, since a prediction needs to both be correct in terms of its class and ground the object as the correct location with an adequate spatial precision. We measure the spatial overlap between a predicted bounding box and the true "ground-truth" box with intersection over union, or IoU, often defined as: where B_p is the bounding box of the predicted object, and B_g the ground-truth box.

$$IoU = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}$$

Where the numerator is the Area of overlapping of two boxes, and the denominator is the entire area covered by two boxes.

Precision is the fraction of positive detections that are correct, and recall is the fraction of reference objects that are detected. Average Precision (AP) summarizes the precision-recall curve for a given class and means AP (mAP) takes the mean over all classes. These metrics are well established through object-detection benchmarks such as PASCAL VOC and Microsoft COCO [11], [12].

For this study two mAP scores are especially relevant. MAP@50 considers detections at an IoU threshold of 0.50, while mAP@50-95 considers an average across the range 0.50-0.95 in 0.05 steps, making the latter a more challenging metric since the predictions are required to be robust to tighter localization criteria [11]. As downstream canopy analysis relies on the ability to detect tree instances with a good degree of localization, these metrics should be appropriate for this component.

2.2.4 Colour vegetation indices and morphology

While object detection yields the position of each individual tree, a rectangle that is larger than the actual canopy is not a useful metric of canopy-covered area. The canopy shapes are irregular and a bounding box will often contain a significant amount of soil, shadow, background vegetation and open space between the leaves.

The object detection output must therefore be processed further to enable reliable estimates of image-area usage.

Colour vegetation indices afford a fast way to differentiate vegetation in colour RGB images. Woebbecke et al. tested a suite of colour indices and their applicability in separating plant material from soil and crop residue and found the Excess Green (ExG) principle very helpful [5]. Meyer and Neto later applied colour vegetation indices to automatic crop images and found the EXG methodology useful to separate the vegetation from the background in traditional colour images [9]. In this work the ExG principle is applied as.

$$ExG = 2G - R - B$$

Where R, G, and B are the red, green, and blue pixel-channel values. Pixels that have stronger responses in the green-channel are assigned higher ExG values.

ExG's primary benefits for this study are its simplicity, interpretability and the fact that it does not require a separately obtained segmented data set. On the other hand, colour based separation of vegetation can be affected by lighting conditions, shadows, fading or pale leaves and other green objects in the scene. Therefore, the ExG component is limited to the areas of the tree detections, rather than being unconstrained over the entire aerial image. Two binary morphological operations are then applied to improve the resulting mask, namely closing and opening. Closing can fill small holes and gaps within connected areas of vegetation, while opening can remove small dots of foreground noise [10]. This provides an efficient means to produce an improved approximation of the canopy occupancy from the raw colour-based mask.

2.2.5 Distance transforms and free space

Once the occupied vegetation has been generated as a binary mask, the rest of the space can be regarded as free space. By itself, the number of free pixels is not an indicator of whether the free space can allow for another planting position as a high percentage of free pixels could mean large numbers of thin strips of space in between

canopies. Spatial clearance should be considered as well as total free area. A Euclidean Distance Transform (EDT) creates for each free pixel a value corresponding to the Euclidean distance between that pixel and the closest occupied pixel. Maurer, Qi, and Raghavan introduced an efficient exact Euclidean distance-transform algorithm for multidimensional binary images [6]. If one applies this to the veg- etation occupied/free binary mask in this study, the Euclidean Distance Transform effectively produces a continuous clearance surface from the mask of free space. Pixels in the vicinity of a canopy will be assigned small values, and pixels in the vicinity of the centre of large open regions will be assigned larger values. Local maxima in this clearance surface can be taken as points that are somewhat further from existing vegetation and can therefore be used as candidate open space centres. This is important for planting-capacity estimation because it separates 'having free area' from 'having a large enough free area'. This component then applies this principle of free pixels separated by a patch of free pixels to an iterative rule for spacing, so that candidate points can be converted into an approximate planting-capacity estimate, rather than treating all free pixels as of equal value.

2.3 Related research

Several works have explored the use of deep learning and aerial imagery for automated palm-tree detection, counting, and inventory production. Mubin et al. [8] proposed a deep-learning-based approach to detect and count young and mature oil palms, demonstrating the ability of deep learning approaches to identify individual palms across different growth stages. Culman et al. [13] similarly proposed a deep-learning approach to detect individual Phoenix palms from RGB remote-sensing images and demonstrated its potential to provide large-scale inventories of trees. These work showcase how computer vision can dramatically reduce the need for manual tree counting and also offer spatial information that could be used for monitoring plantations, but their main contribution in terms of data is in the form of detections, counts, or inventories of trees, not necessarily a quantification of remaining capacity.

A notable and relevant study to our data was performed by Ammar et al., [7] in which they designed an automated palm-tree detection, count, and geolocate process

based on aerially acquired geotagged imagery across large farm areas. They evaluated a number of modern deep-learning object detector architectures (Faster R-CNN, YOLOv3, YOLOv4, and Efficient Det) for this purpose, showing their suitability for the Palm and other-tree classes in the present land-utilization environment. The aerial imagery used in the public Palm/Tree dataset-used here as an input to the detector component-was provided with the Ammar et al., [7] study; thus, the present study takes a known aerial palm-detection problem and further exploits the detected objects as navigation inputs, spatial cues, and generation inputs to further analysis of land-utilization, vegetation-occupancy, free-space, and approximate planting-capacity.

Additional research by Chowdhury et al. [14] for oil-palm counting from drone images identified real-world challenges of variability in image viewpoints, oblique angle imaging, mixed vegetation, variable crown shape, and tree overlap. Deep-learning-based detection used by these authors is shown to partially compensate for difficult conditions when counting palms in aerial images. While this analysis is carried out at the tree level, these same issues of variable canopy shape and occlusion will impact the accuracy of the land-occupancy estimation, as a rectangular detection box for each tree can, at best, be a poor estimate of its canopy footprint.

Thus far, previous palm-detection studies have demonstrated that combination of aerial imagery with deep learning can be utilized for preeminent tree localization and counting [7], [8], [13], [14]. Despite these achievements, the approach presented here contrasts with these studies, given our stated goal of preservation of detection as an intermediate step rather than the end result. After detecting Palm and Tree instances, the vegetation assessment based on the ExG is then used to estimate canopy occupancy, and using a Euclidean Distance Transform, identify regions of high-clearance open space; then, a spacing-aware greedy procedure estimates an approximate number and distribution of future planting slots from this free-space estimate. This transition from detection canopy occupancy spatial clearance approximate capacity is the key distinction of this work relative to prior detection, inventory, counting, and geolocation studies.

The literature surveyed suggests that to-date, no individual approach can fully solve the land-utilization issue. Object detectors yield accurate object positions but not detailed canopy outlines; colour-index techniques can estimate the nature of the vegetation without having to annotate segmentations (although they are susceptible to illumination and colour of the foliage); distance transforms can determine local clearance but a informative occupy/free-space. Therefore this project takes a hybrid and sequential approach, utilizing these existing techniques in accordance with their advantages rather than seeking to substitute them with an end-to-end model.

2.4 Existing systems products and techniques

Three broad approaches exist for the land-use question addressed here (Table 2.1).

Table 2.1: Approaches to estimating tree occupancy and free space from aerial imagery

Approach	Strengths	Weaknesses for this problem
Manual interpretation of aerial images or field survey	Simple; incorporates expert judgement	Slow, subjective, hard to repeat across sites and dates
End-to-end semantic or instance segmentation	Delineates canopy pixels directly	Needs pixel-level training masks, which were not available
Hybrid: object detection + colour-index mask + spatial analysis (this work)	Uses box labels only; each stage interpretable and replaceable	Mask accuracy depends on colour threshold; capacity in pixels unless GSD known

There are several ways to analyze the distribution of trees and the areas available for planting, using aerial photography. The simplest is a form of visual interpretation where plantation managers or field officers look for trees and available areas in the plantation or aerial photographs, but it is time consuming, subject to bias and not repeatable. For this reason, automated image-analysis is becoming popular for repeatable assessment of the distribution of trees and the use of land for cultivation.

A more sophisticated method is semantic/instance segmentation where pixels are classified to segment out object area or tree canopies. Deep-learning based semantic segmentation models like U-Net [15] can successfully detect object areas at pixel resolution. This could, in theory, produce good canopy boundaries, but usually needs pixel level 'ground-truth' mask for training and evaluation of the model. The aerial Palm/Tree dataset used here has bounding-box labels rather than hand-crafted

canopy masks, so training and developing a separate segmentation model was not deemed appropriate for this dataset and project.

There has been a vast range of both two- and single-stage object detectors used for tree localization in computer-vision studies; for example, Faster R-CNN [16] and YOLO. Ammar et al. [7] has tested a range of deep-learning object detectors for aerial palm-tree detection and found that these techniques are capable of robust palm identification and counting at aerial scales. After these prior test results, and considering the specific needs for this component, YOLO11n was chosen as the tree-detection method as it is capable of outputting a of a small pretrained model, and providing the bounding boxes, classes and confidence values for input to later stages in the land analysis pipeline. This study makes no general claims that YOLO11n is the best network architecture, as no side-by-side comparison to other detectors was performed.

Various options were considered for estimating occupied land area. Simply accepting the full bounding-box area around an object is the easiest option but severely overestimates actual occupied area in cases of diffuse irregular canopies, soil, shadows and background pixels. First, simple geometric approximation based on the use of detected centers of trees, combined with an average canopy size metric, was therefore implemented as a circular-buffer estimate. Circular buffers do assume a relatively regular canopy shape and can overestimate occupied areas, since bare ground may be included. The final estimate uses an Excess Green vegetation mask within our detection regions, avoiding the creation of a dedicated segmentation model, in a relatively straightforward way that is more image responsive.

Various alternative approaches can also be taken to the rest of the free-space analysis. Reporting only the percentage of available pixels is inadequate, as it ignores the sizes and distribution of the gaps; dividing the total available area by an average canopy size has the same problems and can also lead to an overestimate when the free space is broken into multiple long, narrow fields. This thesis therefore chooses to use a Euclidean Distance Transform to determine the degrees of openness of free pixels to the existing vegetation; these degrees of openness are then used by a greedy placement routine which iteratively chooses the most open position while ensuring that all subsequent proposed positions do not violate that same spacing constraint.

In general, the retrieved design integrates object detection, vegetation-based occupancy filter, and free-space filter based on distance. This hybrid design is suitable for the available data since object-level annotations exist, but pixel-level canopy masks and real-world GSD do not exist. In addition, the intermediate results of the build are retained, thus making it possible to analyze the detections, vegetation masks, free-space masks, and proposed planting points individually.

2.5 Gap synthesis and conceptual framework

The individual techniques used for this work which are described in the previous sections, are thus known. Deep-learning object detectors are capable of locating and counting palms from overhead imagery [7], colour vegetation indices such as Excess Green can be used to separate vegetation from the background [5], [9] and Euclidean Distance Transform algorithms can measure the degree of spatial freedom in free areas [6]. These techniques can be considered to be well established and have been used in isolation, but not on their own can they give a complete solution to the land-use problem tackled in this work. They can give an estimate of the number and approximate locations of existing trees, but not of actual canopy coverage [7]; a vegetation mask indicates occupancy but not whether free space is of a large enough area or of appropriate shape to accommodate further plantations [5], [9]; and simply calculating a percentage of free area does not take fragmentation or minimum spacing into account [6].

The knowledge gap, therefore, is access to a transparent workflow that can turn Palm/Tree aerial instance detection into occupancy area estimates, and free-space estimates, and a proxy for planting capacity. This is what this work provides through the following 4 processing steps in order:

1. Detect Palm and Tree instances from the aerial RGB image using YOLO11n and generate their bounding boxes, confidence scores, classes, and spatial locations.

2. Estimate vegetation occupancy by first using a circular-buffer baseline and then refining the occupied area with an ExG-based vegetation mask combined with elliptical constraints and morphological processing.

3. Analyze the remaining free-space mask using the Euclidean Distance Transform and local-maxima detection to generate a clearance surface and identify ranked high-clearance candidate pockets.

4. Estimate approximate additional planting capacity using a greedy placement procedure that selects proposed planting positions while enforcing an image-space spacing constraint.

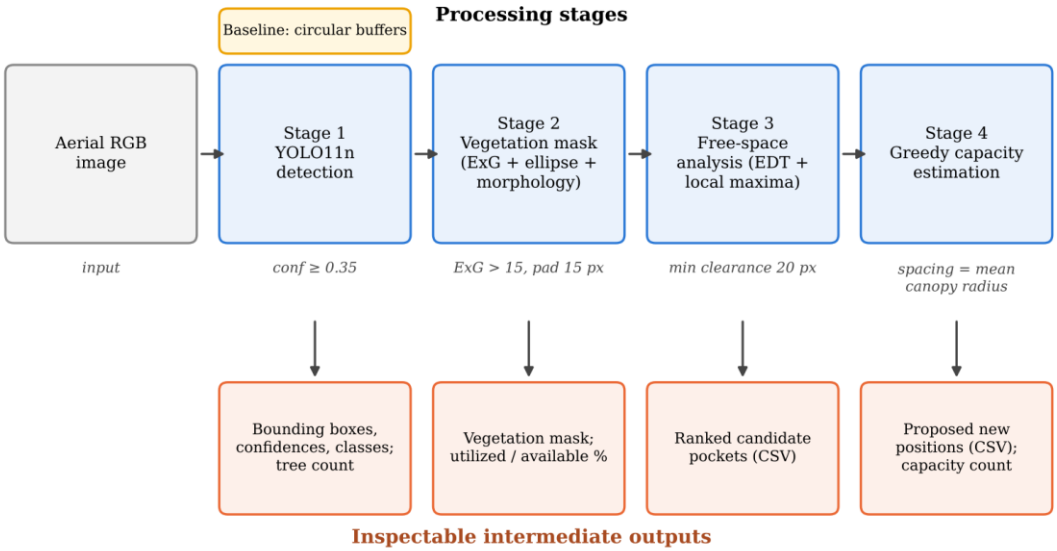


Figure 2.1: Conceptual framework of the land-utilization component, showing the four processing stages, their key parameters and the inspectable output of each stage

Hence, a circular-buffer occupancy estimate is maintained as a baseline, and the weed-optimized vegetation mask yields a somewhat more image-sensitive estimate of visible canopy cover. Keeping the stages separate is also more amenable to interpretation as each of the detection, vegetation masking, free-space detection, and capacity estimate output images can be viewed in isolation. Finally, it will be easier to improve individual stages going forward, for example using a canopy-segmentation trained mask instead of the ExG mask, and the pixel-based spacing rule of thumb replaced with a metric agronomic spacing standard, once Ground Sampling Distance is available.

3 RESEARCH METHODOLOGY

3.1 Research design

The work presented here uses a quantitative experimental software-research paradigm to design and to implement and evaluate the proposed aerial-land-utilization-and- approximate-planting-capacity-analysis component. The experiment is designed around two evaluation levels. At the model level the YOLO11n object detector is trained, and its performance is evaluated with a labelled validation data set to determine how well it can localize, and to establish the number of Palm and Tree objects that it can identify. Metrics used for this evaluation are the standard computer vision object-detection metrics of precision, recall, mAP@50, and mAP@50-95. At the pipeline level the trained detector is used to localize Palm and Tree instances within the high-resolution aerial image, and the analysis pipeline is completed to produce output estimates of canopy occupancy, free-space candidates, and approximate additional planting sites.

The development was performed as an iterative refinement process. The primitive method simply used a circular canopy buffer to represent each detected tree and derived a coarse measurement of the occupied image area. Visual examination of this simple result revealed that circular areas would often include large amounts of exposed ground between loosely dispersed palm fronds, so the method was improved by applying an ExG mask within detected regions, further refined via morphology, then subsequently analysed by the Euclidean Distance Transform. Again, because the image area available could not, in itself, determine if it was dense enough for more planting, an additional greedy, spacing-aware placement stage translated spatial clearance into a coarse estimate of planting capacity.

The approach makes it possible to assess and interpret each of the stages of the component independently while keeping an explicit connection between research questions and methods. The detector is concerned with detecting and locating existing trees, the vegetation-analysis stage with the canopy-occupancy computation, the distance-transform stage with the assessment of the land's spatial structure, and the greedy placement stage with estimating additional planting capacity under an

image-space spacing constraint. The methodology thus relies on a mixture of deep-learning experimentation and deterministic image-processing and spatial-analysis techniques that yield transparent and reproducible land-uses results.

3.2 Data participants or study materials

This work did not involve any human participants. Palm Dataset The publicly accessible Aerial Images of Palm Trees dataset was used as the base for detector development, which was used in conjunction with the aerial palm-counting and geolocation work undertaken by Ammar et al [7]. The collection of the dataset was accessed via Kaggle under the Palm-Counting-349images collection. The detector training set consisted of 279 labelled aerial images containing 10,398 total bounding-box annotations. The original annotation data file train_labels.csv contained the following columns: filename, width, height, class, xmin, ymin, xmax, ymax. The implemented object detector was trained to two classes, Palm (class 0) and Tree (class 1).

The splitting of the dataset was done at the image level rather than at the bounding-box level to ensure that all annotation of a given aerial image resided within the same partition. A stratified random split with testsize = 0.15, randomstate = 42 resulted in 237 training images and 42 validation images. This stratified split minimised the probability of information leakage to occur if the same image's annotations feature in both the training and validation sets, but achieved comparable sizes of both dataset splits as the purely random split. From 279 annotated images, all were present with no missing or corrupt files. The validation split held 1616 objects, with 1357 Palm instances and 259 Tree instances, demonstrating a class imbalance with the Palm class being dominant.

Further, one high-resolution aerial image (from the shared test folder of the dataset, image name: ck2euxc9kxgvm07486g2d5pid.jpg, resolution 4864 × 3648) was randomly chosen to illustrate the entire land-utilization workflow in addition to the training and validation datasets. It was not used for training the detector, nor was it included in the measurement of the detector-validation performance. It was only used for the subsequent demonstration of tree detection, canopy-occupancy calculation,

free-space assessment, and planting-capacity estimation. Since the source image was not from Sri Lankan coconut gene banks but from an arid palm-farm environment, it was considered a proxy data set for demonstration of the proof-of-method.

Table 3.1: Dataset and split used for detector development

Item	Value
Source	Aerial Images of Palm Trees (Kaggle, RIOTU) [4]
Annotated training images used	279
Bounding boxes	10,398
Classes	Palm (0), Tree (1)
Split method	Image-level random split, test_size = 0.15, random_state = 42
Training images	237
Validation images / instances	42 / 1,616 (Palm 1,357; Tree 259)
Missing or corrupt images	0
Demonstration image	test/ck2euxc9kxgvm07486g2d5pid.jpg, 4864 × 3648 px

Another (single) image from the separate test folder of the dataset was employed to illustrate the entire land-analysis workflow. This single source image was never shown to the detector during model training and was not employed for metrics on detector validation. The source image is of palm-farm land outside of Sri Lanka, so considered as a stand-in for proof-of-method rather than a real sample of the targeted Sri Lankan coconut gene-bank environment.

3.3 Experimental or study procedure

The overall experimental procedure was designed as a sequential pipeline for converting the base aerial-image annotations into a trained Palm/Tree detector, and then converting the Palm/Tree detector outputs into land-utilization information and estimated planting-capacity scores. The six sub-stages involved in the overall workflow are annotation conversion and dataset preparation, YOLO11n fine-tuning, inference and baseline occupancy estimation, fine-grained vegetation-mask refinement, free-space candidate detection, and greedy approximate capacity

estimation. The output of each sub-stage is used as the input for the next, which makes the overall pipeline transparent and independently verifiable.

3.3.1 Annotation conversion and dataset preparation

Initial dataset annotation was in the form of CSV with bounding-box corner coordinates using xmin, ymin, xmax, ymax. These were converted into center-based coordinates that are formatted for training the YOLO11n. For each image of width W and height H, the normalized centres point coordinates, x_c and y_c, were derived from:

$$x_c = \frac{x_{min} + x_{max}}{2W} \qquad y_c = \frac{y_{min} + y_{max}}{2H} \tag{3.1}$$

The normalized bounding-box width \(w\) and height \(h\) were calculated as:

$$w = \frac{x_{max} - x_{min}}{W} \qquad h = \frac{y_{max} - y_{min}}{H} \tag{3.2}$$

The above calculations convert the absolute pixel coordinate locations to x and y based on the width and height of each image respectively, which keeps the annotation valid when the images are scaled as part of model training. Each annotation was then saved as class x_c y_c w h in the YOLO label format and saved in individual text files of the same name as the respective aerial image. Palm = 0, Tree = 1.

After conversion the images and label files were sorted into train and validation directories as defined in the image-level split described earlier. The dataset was prepared into images/train, images/val and their corresponding label directories as a result. Additionally, a data.yaml configuration file was created with dataset location, training and validation paths, number of classes (nc = 2) and the classes' names (Palm, Tree) that was ready to be used in the Ultralytics YOLO11 training framework.

3.3.2 YOLO11n fine-tuning

A pretrained YOLO11n model was fine-tuned for the Palm/Tree aerial object-detection task within the Ultralytics framework [4]. The COCO-pretrained yolo11n.pt checkpoint was selected as the starting point, with the intention of leveraging the visual features learned during generic object detection rather than training the network from scratch with random initialization. When initializing the model, 448 out of 499 weight tensors were transferred from the pretrained checkpoint, while the class-specific detection head was adapted for the two target classes, Palm and Tree.

Training was carried out for 50 epochs with an input size of 640 × 640 pixels and batch size of 8. The training included an iterative process of about 30 iterations per epoch and was trained on a single NVIDIA Tesla T4 GPU with Automatic Mixed Precision. Early stopping patience was set to 15 epochs; however, the training lasted for the 50 epochs as the stopping criterion was not obtained.

The optimizer was set in the training framework (Ultralytics) to adamw with an initial learning rate of 0.001667, momentum of 0.9, and weight decay of 0.0005. A three-epoch warm-up period was applied. Several augmentation strategies were employed in training to help the detector generalize to variations in aerial images, these were: Mosaic augmentation, horizontal flip (p=0.5), HSV colour augmentation, translation, scale, random erasing, and Rand Augment. Mosaic augmentation was disabled in the last 10 epochs to let the detector home in on its predictions based on a more natural image composition.

Training was deterministic with seed=0 to maximize reproducibility. Total training time (over 50 epochs) was around 0.209 hr (or 12.5 min). Model checkpoints were monitored during training, and then the best checkpoint was chosen for its determined Ultralytics fitness based on the detector's performance relative to mAP, and then used for the validation, demo image inference, downstream land-utilization analysis steps (started with the best checkpoint at epoch 39).

Table 3.2: YOLO11n training configuration (from the notebook run log)

Parameter	Value
Initial weights	yolo11n.pt (COCO-pretrained); 448/499 tensors transferred
Epochs / patience	50 / 15 (early stopping not triggered)
Image size / batch	640 × 640 / 8
Optimizer	AdamW (auto): lr 0.001667, momentum 0.9, weight decay 0.0005
Warm-up	3 epochs
Augmentation	Mosaic (off last 10 epochs), flip-LR 0.5, HSV, translate 0.1, scale 0.5, erasing 0.4, RandAugment
Precision / determinism	AMP enabled; deterministic, seed 0
Hardware / time	NVIDIA Tesla T4 (14,912 MiB); 0.209 h for 50 epochs
Checkpoint selection	Best fitness = 0.1 × mAP@50 + 0.9 × mAP@50–95 (epoch 39)

3.3.3 Inference and circular-buffer baseline

Once the model training and validation had been completed, the optimal checkpoint of YOLO11n was used on the separate high-resolution demo image. Inference was carried out with a confidence score threshold of 0.35, with detections falling below this threshold not considered in the land-analysis step. For each of the detected Palm/Tree objects, the bounding-box coordinates, predicted class and confidence score and object center point were obtained. This was used as the spatial reference point for the first land-occupancy estimation.

Since the actual box itself contains a large amount of background information around irregularly shaped trees, the boxes were not summed to estimate occupied area. Instead, a circular approximation to the canopy was used around each center detection, which was based on the mean width and mean height of detected boxes, with a common canopy radius r given by:

$$r = \frac{\text{mean}(w_{box}) + \text{mean}(h_{box})}{4} \tag{3.3}$$

Where w_{box} and h_{box} are the widths and heights of the bounding boxes for the detected objects in pixels. This provides a geometrical approximation of the typical radius of the canopy, depending on the image, instead of a manually selected fixed radius.

Next, circular buffers with a radius of r were created around all tree centres found. Overlapping buffers were unioned together and cut to the limits of the image (such that regions outside the image were excluded). The area occupied by the set of circular buffers was considered baseline used image area and the amount of space left over baseline available area. This circular-buffer approach was preserved as a straightforward geometric baseline estimate of vegetation-mask estimate, made later using the ExG.

The locations of detected trees were used to create a separate Gaussian Kernel Density Estimate to graph the spatial frequency of the trees. The bandwidth of this layer was 2% of the longer side of the image. This heat map was only used for visualization purposes and was not input into the utilization, free-space or planting-capacity analyses.

3.3.4 Refined vegetation mask

The circular-buffer technique only gave a rough estimate of canopy occupancy since all detected trees were assigned to occupy a perfect circle area. To derive a more image-sensitive estimate, the RGB demonstration image was processed with an ExG vegetation index. The image was converted into floating-point range (0-255) values, then for each pixel, the index value was derived from:

$$ExG = 2G - R - B \tag{3.4}$$

Where R, G, and B are the red, green, and blue channels, respectively. Pixels with the higher ExG should be regions where the green channel is high – and therefore the

index is appropriate for detecting visible vegetation in the identified tree regions. For each YOLO11n detection, the original bounding box was padded by 15 pixels in all directions (when not truncated by the edge of the image). This extra padding was to allow the detection region to encompass palm fronds that might extend outside the bounding box but not extend beyond the image. A pixel was classified as vegetation if it had an ExG value > 15 and was contained within an ellipse fitted within the padded detection region. This elliptical constraint reduced the effect of green and shadow pixels near the corners of the detection box, and produced a more circular, canopy representation that better matched the known relatively radial structure of the crowns.

Detection mask derived vegetation masks were then combined through a logical-OR operation to generate the full-image binary canopy mask. Morphological processing was then used to eliminate unlikely small artifacts. Through the use of a 5 × 5 binary closing operation on the detection mask, small holes in detected vegetation areas were filled, while subsequent 3 × 3 binary opening operations eliminated isolated pixels and unwanted small regions of noise. Since ExG processing was only done in a region around each YOLO detection and not across the entire aerial image, many green groundcover areas far from detected trees that would otherwise have contributed to the estimate of occupied area were excluded.

The final refined vegetation mask was then used to compute used and available percentage of image area, and was also used as the binary occupancy input for the free-space analysis. The parameters chosen for the box padding, ExG threshold, morphological kernels and free-space processing were implementation parameters for this pipeline example that were not learned.

Table 3.3: Vegetation-mask and free-space parameters

Parameter	Value	Purpose
BOX_PADDING_PX	15 px	Capture fronds extending beyond the box
EXCESS_GREEN_THRESHOLD	15	Minimum ExG for a vegetation pixel
USE_ELLIPSE_CONSTRAINT / ELLIPSE_SCALE	True / 1.0	Suppress corner artefacts
Closing / opening structure	5 × 5 / 3 × 3	Fill holes / remove specks
GRID_DOWNSAMPLE	2	Halve resolution for distance computations
MIN_DISTANCE_BETWEEN_POCKETS	15 grid cells (≈ 30 px)	Separation between candidate peaks
MIN_CLEARANCE_PX	20 px	Minimum clearance of a candidate pocket
REQUIRED_SPACING_PX	Auto = r = 148.2 px	Spacing rule in greedy placement
MAX_NEW_TREES_TO_FIND	500	Safety cap on the greedy loop

3.3.5 Free-space candidate detection

Once the result vegetation mask was created, the mask was inverted such that pixels not classified as vegetation were mapped as free space. To perform distance analysis, since the demonstration image has a resolution of 4864 × 3648 pixels (17.7 million pixels), the binary free-space mask was down-sampled by a factor of 2 to reduce computational cost of the distance analysis. An exact Euclidean Distance Transform (EDT) was then performed on the down-sampled free-space mask. Distance values were multiplied by 2 so that all the reported clearance values are defined in the original image scale.

The free pixels of the EDT are each ascribed a distance value that indicates the shortest distance to an occupied vegetation pixel. As such, the closer a pixel is to an established canopy, the lower its clearance value; whilst pixels in wider open areas and towards the centre of open areas receive higher values. This distance surface was then used to identify ecologically important open areas rather than not treat all available pixels equally.

Candidate free-space pockets were then automatically located by running a maximum local filter over the distance-transform surface. To avoid having candidate peaks located very close to one another, candidates were only accepted if they were at least 15 cells apart in the down-sampled grid (about 30 pixels full resolution). As another check we also ignored candidate points with an absolute clearance of less than 20 pixels, to avoid having a candidate located in a tiny open region. A list of the acceptable candidate points was maintained, with each candidate having stored the pixel location and the clearance. The list was then sorted in order of decreasing distance, so the most open spaces could be considered first.

The candidate locations were written to the file plantingcandidatespots.csv for further analysis and plotting. Importantly, these candidate pockets are not themselves candidates for final capacity due to the possibility of multiple local maxima within a single large free-space region. Instead, they give an intermediate image representation of the most widely spaced out regions of the image and are consumed by a greedy capacity-estimation step.

3.3.6 Greedy approximate capacity estimation

The candidate free-space pockets identified through local-maximum analysis cannot be interpreted directly as additional planting capacity because a single large open region may contain several nearby clearance peaks. Counting every peak as an independent planting position could therefore overestimate the number of trees that can be accommodated. To address this issue, a greedy spacing-aware placement procedure was applied as the final stage of the land-analysis pipeline.

The procedure begins with the down-sampled refined canopy mask treated as a blocked grid. At each iteration, the Euclidean Distance Transform is recalculated over the currently available space, and the free location with the greatest clearance from the blocked area is selected. A location is accepted only when its clearance is greater than or equal to the required spacing value \(S\). Once a position has been accepted, a circular region of radius \(S\) around that position is added to the blocked grid. The distance transform is then recalculated and the process is repeated. This iterative update prevents subsequent planting positions from being placed too close to previously accepted locations.

The algorithm terminates when no remaining free location has a clearance value of at least \(S\), or when the predefined safety limit of 500 proposed positions is reached. In the current implementation, no real-world Ground Sampling Distance was available; therefore, the required spacing was automatically derived from the previously calculated mean canopy radius,

$$S = r = 148.2 \text{ pixels} \quad (3.5)$$

3.4 Measures and evaluation metrics

Detector metrics. The detector was evaluated on the 42-image validation split with the standard Ultralytics validator:

$$\text{Precision} = TP / (TP + FP), \quad \text{Recall} = TP / (TP + FN) \quad (3.5)$$

$$\text{IoU} = \text{area}(B_{pred} \cap B_{gt}) / \text{area}(B_{pred} \cup B_{gt}) \quad (3.6)$$

AP is the area under the precision–recall curve of one class at a given IoU threshold; mAP@50 averages AP over the two classes at IoU = 0.50, and mAP@50–95 additionally averages over IoU thresholds 0.50:0.05:0.95 [7], [6]. These are detection metrics and must not be read as classification accuracy. The checkpoint with the highest fitness, 0.1 × mAP@50 + 0.9 × mAP@50–95, was kept as best.pt.

Land-analysis measures. Utilization was computed from the cleaned mask as

$$\text{Utilized (\%)} = 100 \times N_{vegetation} / N_{total} \quad (3.7)$$

$$\text{Available (\%)} = 100 - \text{Utilized (\%)} \quad (3.8)$$

and, for the baseline, as the area of the merged buffer polygon divided by the image area. Free-space quality is described by the number of candidate pockets and their clearance in pixels. Capacity is the number of positions accepted by the greedy algorithm under spacing S. All land measures are in image space.

3.5 Data analysis

Detector results were analysed overall and per class, because the two classes differ greatly in frequency and appearance and an overall score can hide a weak class. Per-epoch training losses and validation metrics were extracted from the run log to examine convergence and to identify the best epoch (Figure 6.1).

The land-analysis stages were analysed as successive approximations rather than competing models. For the demonstration image the following were recorded and compared: detection count and confidence range; baseline versus refined utilization; number and clearance of candidate pockets; and greedy capacity. Visual sanity-check panels (original image, binary mask, free-space overlay) were used to judge qualitatively whether the mask followed the visible canopies, since no canopy ground truth was available for a quantitative comparison.

3.6 Ethics privacy security and responsible AI

The detector was developed on a public agricultural image dataset that contains no personal data, and the component performs no surveillance, authentication or profiling. The main ethical concern is responsible interpretation of automated outputs. The 55-position figure in particular must not be presented as a verified planting plan, because the analysis contains no ground scale, soil, terrain, irrigation, access or expert input. Results are therefore always reported with their unit (pixels) and their assumptions.

The dataset was captured outside Sri Lanka and is not coconut-specific, so proof-of-method performance is clearly distinguished from local operational validity. Any future deployment should use locally collected aerial imagery captured with the landowner's permission and in line with national drone regulations, document data provenance, and have planting suggestions reviewed by agronomists before use. Dataset and software licences and attribution requirements must also be respected.

3.7 Reproducibility and research artefacts

Reproducibility is supported by a fixed split seed (random_state = 42), deterministic training (seed 0), explicit configuration constants at the top of every notebook cell,

recorded software versions, and saved model and result files (Table 3.4). The full list of constants is given in Appendix A.

Table 3.4: Key research artefacts produced by the notebook

Artefact	Location / description
Dataset configuration	/kaggle/working/yolo_dataset/data.yaml
Best detector weights	/kaggle/working/runs/palm_detector/weights/best.pt
Exported detector	/kaggle/working/palm_detector_best.pt (5.2 MB)
Candidate pockets	planting_candidate_spots.csv (x, y, clearance_px)
Proposed positions	real_planting_capacity.csv (x, y, clearance_px)
Figures	test_detections.png, test_heatmap.png, empty_spaces_accurate.png, mask_sanity_check.png, real_planting_capacity.png
Source notebook	coconut-land-detection.ipynb (Kaggle)

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Requirements were derived from the aim of turning aerial imagery into interpretable land-use information for non-specialist users (Table 4.1).

Table 4.1: Functional and non-functional requirements

ID	Requirement	Type
FR1	Accept an aerial RGB image for analysis.	Functional
FR2	Detect Palm/Tree instances and return boxes, classes and confidences.	Functional
FR3	Estimate utilized and available image area (baseline and refined).	Functional
FR4	Identify and rank high-clearance free-space pockets.	Functional
FR5	Estimate additional planting positions under a minimum-spacing rule.	Functional
FR6	Export coordinates as CSV and annotated result images.	Functional
NFR1	Use a compact detector suitable for a modest inference server.	Non-functional
NFR2	Keep every intermediate output inspectable.	Non-functional
NFR3	Report pixel-space results with their unit and assumptions; never as metric field recommendations.	Non-functional
NFR4	Be callable from the group platform's backend.	Non-functional

4.2 Component architecture within the complete solution

The component is one of four analytical modules behind the group's common web platform (Figure 4.1). According to the group project design, the platform uses a React frontend, a Node.js/Express backend and a MySQL database, with Python model services exposed through Flask. The notebook explicitly identifies the exported best.pt file as the artefact required by the Flask API. The notebook itself contains the model and analysis pipeline only; the web-stack code is outside its scope, so the platform is described here as integration context rather than as tested evidence.

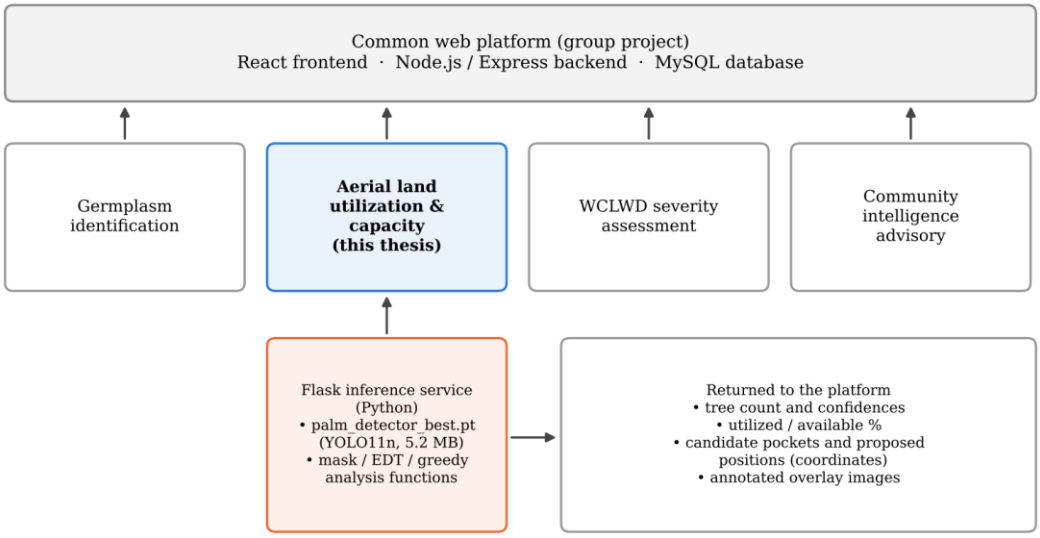


Figure 4.1: Position of the land-utilization component within the complete solution

Inside the component, detection is separated from post-processing. The detector can therefore be retrained or replaced without touching the mask and capacity code, and the mask stage can be upgraded to a learned segmentation model without retraining the detector.

4.3 Design alternatives and decisions

Table 4.2 summarises the main design decisions, the alternatives considered and the reason for each choice.

Table 4.2: Main design decisions

Decision	Alternatives considered	Choice and reason
Task formulation	Image classification	Object detection: one image contains many trees whose locations are needed
Detector	Faster R-CNN; larger YOLO11 variants	YOLO11n: compact (2.58 M parameters), fast, pretrained, simple export; no controlled size comparison was run
Occupancy	Bounding-box area; circular buffers; trained segmentation	Circular buffers kept as a baseline; ExG mask inside padded, ellipse-constrained boxes chosen because no pixel labels exist
Free space	Total empty %	EDT clearance + local maxima: captures gap size, not just gap area
Capacity	Free area ÷ mean canopy area; peak count	Greedy placement: respects spacing from canopy and between new positions
Spacing unit	Fixed metric spacing	Mean canopy radius (148.2 px) because GSD is unknown

A larger YOLO11 model might localise canopies more tightly, at higher training and inference cost; this was not tested. A trained segmentation network would likely outline canopies better than a colour threshold, but no pixel-level masks were available to train or evaluate one, so ExG offered a practical training-free refinement. Peak detection alone was not used for capacity because neighbouring peaks inside one gap would double-count space; greedy placement fixes this by updating the blocked map after each accepted position.

4.4 Detailed design

The detector returns, for each object, a class index, a confidence and box coordinates (x1, y1, x2, y2). The baseline branch uses only box centres and the mean box size. The refined branch uses each box as a region of interest for pixel analysis, which keeps unrelated green ground cover from dominating the occupancy estimate.

The free-space module treats the refined mask as a blocked binary grid at half resolution and rescales distances to full resolution. The capacity module runs the greedy loop below on the same grid. For a given mask and spacing value the output is deterministic.

```
Algorithm 1: Greedy approximate capacity estimation
Input : canopy mask M (True = occupied), spacing S (px), downsample d,
cap K = 500
Output: list P of proposed positions (x, y, clearance)
blocked <- M[::d, ::d];  P <- []
repeat up to K times:
    D <- EDT(not blocked) * d     # clearance in full-res px
    if max(D) < S: stop
    (x, y) <- full-resolution coordinates of argmax(D)
    append (x, y, max(D)) to P
    blocked[cells within distance S of (x, y)] <- True
return P
```

Two properties of the design are worth noting. First, the image border is not treated as an obstacle: the EDT measures distance only to blocked pixels, so a position can be proposed right at the image edge even though the land beyond the edge is unknown. Second, because the canopy mask only covers detected trees, any tree the detector misses is treated as free ground. Both effects tend to increase the capacity estimate and are discussed in Chapter 7.

4.5 Standards and professional practice

Professional practice is reflected in reproducible splitting and seeding, strict separation of training and validation images, use of standard COCO-style detection metrics, preservation of model and result artefacts, and explicit reporting of units and limitations. The thesis avoids calling mAP an “accuracy”, separates validation metrics from single-image results, and records the absence of metric scale instead of silently treating pixels as metres. Citations follow the IEEE style, and dataset and software provenance is acknowledged. Code follows a clear step structure with named configuration constants and inline documentation of every parameter's purpose.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The component was developed and run entirely in a Kaggle notebook with GPU acceleration. Table 5.1 lists the technologies used and their roles.

Table 5.1: Development environment and technologies

Technology	Role
Kaggle Notebook	Training and experimentation environment with GPU
Python 3.12.13	Implementation language
Ultralytics 8.4.116 [3]	YOLO11 training, validation and inference
PyTorch 2.10.0 + CUDA 12.8 [18]	Deep-learning backend
NVIDIA Tesla T4 (14,912 MiB)	GPU acceleration
pandas	CSV annotation and result handling
NumPy [19]	Pixel-level array operations
scikit-learn [14]	Image-level split; kernel density for visualisation
Shapely [13]	Circular-buffer geometry and union
SciPy [16]	Binary morphology and Euclidean distance transform
scikit-image [17]	Local-maximum (peak) detection
Matplotlib [20]	Result visualisation

The fused YOLO11n model has 101 layers, 2,582,542 parameters and 6.4 GFLOPs at 640 × 640 input. These figures are given for reproducibility, not as hardware-independent performance claims.

5.2 Component implementation

The notebook is organised into five working steps that correspond to the pipeline stages:

1. **Dataset preparation (cell 1).** Reads train_labels.csv, maps class names to indices, splits images, converts boxes with Equations 3.1–3.2, writes label files and data.yaml, and reports written and skipped images.

2. **Training (cell 3).** Loads yolo11n.pt, calls model.train() with the configuration in Table 3.2, then model.val() on the best checkpoint and prints mAP@50 and mAP@50–95.
3. **Detection and baseline (cell 5).** Calls model.predict() at conf = 0.35, draws boxes, confidences and centres, computes the circular-buffer utilization with Shapely and draws the KDE heat map.
4. **Refined mask and pockets (cell 6).** Builds the ExG mask, cleans it, computes utilization, finds clearance peaks, exports planting_candidate_spots.csv and draws the numbered top-10 map and the three-panel sanity check.
5. **Greedy capacity (cell 7).** Rebuilds the same mask, runs Algorithm 1, exports real_planting_capacity.csv and draws proposed positions over the free-space overlay; a final cell copies best.pt to palm_detector_best.pt for download.

The mask is computed with vectorised NumPy operations and boolean composition, and the analysis consumes the detector's output arrays directly, so no manual hand-over is needed between model and post-processing. Results are held in pandas DataFrames and written to CSV for downstream use by the platform.

5.3 Integration with the other components

The four group components address different analytical tasks with independent datasets and evaluation metrics, and they are combined at application level rather than through a single shared model. The confirmed group-platform architecture uses a React frontend, a Node.js/Express application backend, Flask-based Python model services, and a MySQL database. Within that architecture, this component contributes the exported YOLO11n checkpoint and Python analysis logic, with outputs consisting of detected tree counts, utilized and available image percentages, candidate and proposed pixel coordinates, and annotated result images.

The coconut-land-detection notebook evaluated in this thesis contains the model-development and land-analysis pipeline, but not the complete production routing, database, or deployment code. Consequently, this individual thesis treats the web stack as integration context and reports only artefacts that were directly produced and verified by the notebook. End-to-end API behaviour, request/response contracts,

database persistence, and deployed user-interface behaviour should be documented only from the final group implementation where separate source-code or test evidence is available.

5.4 Verification and software testing

Verification was performed incrementally at each stage by checking printed counts and metrics and by inspecting the generated figures (Table 5.2). Dataset preparation verified the class mapping and confirmed that all 237 training and 42 validation images were written with no missing-image skips. Model validation was performed on the held-out 42-image partition. For the land-analysis stages, the three-panel mask sanity check (Figure 6.5) was used as a qualitative verification aid to inspect whether the ExG mask, ellipse constraint, and morphological operations produced canopy-shaped regions rather than rectangular detection-box artefacts. Because no manually annotated canopy masks were available, this visual check is not treated as quantitative segmentation validation.

Table 5.2: Verification cases and observed results

Test	Expected behaviour	Observed result
Dataset preparation	All split images and labels written	237 train + 42 val images; 0 skipped; 0 corrupt
Class mapping	Two classes detected and mapped	{Palm: 0, Tree: 1}; nc = 2
Training	Loss decreases and metrics rise	Box/cls/DFL loss 1.974/3.437/1.518 → 1.180/0.635/1.043; mAP@50 0.329 → 0.835 (best)
Validation	Metrics produced on held-out split	P 0.782, R 0.824, mAP@50 0.835, mAP@50–95 0.489
Model loading	best.pt loads with correct class names	Loaded; names {0: Palm, 1: Tree}
Inference	Multiple detections on test image	86 detections, confidence 0.35–0.90
Baseline occupancy	Percentages sum to 100	32.29% + 67.71%
Refined occupancy	Percentages sum to 100; mask follows canopies	23.81% + 76.19%; confirmed visually
Candidate pockets	Ranked pockets above 20 px clearance	172 pockets; top clearance 388.4 px
Greedy capacity	Loop stops when no cell has clearance ≥ S	55 positions at S = 148.2 px (cap of 500 not reached)
Export	Model and CSV/PNG files written	All files present; palm_detector_best.pt 5.2 MB

Formal unit tests and automated regression tests were not written in the notebook; they are recommended once the analysis functions are moved into the platform's service code.

5.5 Deployment and operation

The best checkpoint was copied to /kaggle/working/palm_detector_best.pt for download (5.2 MB, reported by Ultralytics as 5.5 MB in decimal units after optimizer stripping). This compact file can be loaded by any Python service with the Ultralytics package. During validation on the T4 GPU the model needed about 3.9 ms of inference per 640-pixel image (0.2 ms pre-processing, 3.5 ms post-processing); a separate validation call reported higher per-image times (35.9 ms inference) that

include start-up overhead, so latency on the deployment server must be measured separately. The analysis stages depend only on standard scientific Python libraries.

Operational use requires the same thresholds and parameters as in Table 3.3 to reproduce the reported behaviour. The component should be regarded as a prototype: before field use it needs images with known GSD, a verified spacing rule expressed in metres, plot-boundary masks and evaluation on locally annotated Sri Lankan coconut imagery.

6 RESULTS AND VALIDATION

6.1 Research results

6.1.1 Training behaviour

Figure 6.1 shows that all three training losses decreased across the run: box loss from 1.974 to 1.180, classification loss from 3.437 to 0.635, and distribution-focal loss from 1.518 to 1.043. Validation performance improved rapidly during the early epochs and then fluctuated around a plateau. mAP@50 first exceeded 0.80 at epoch 20 and reached its highest recorded value of 0.835 at epoch 39, where mAP@50–95 was 0.489. Epoch 39 therefore corresponds to the best checkpoint used for subsequent validation and inference. At epoch 50, mAP@50 was 0.826 and mAP@50–95 was 0.481, confirming that the last checkpoint was not the best-performing checkpoint.

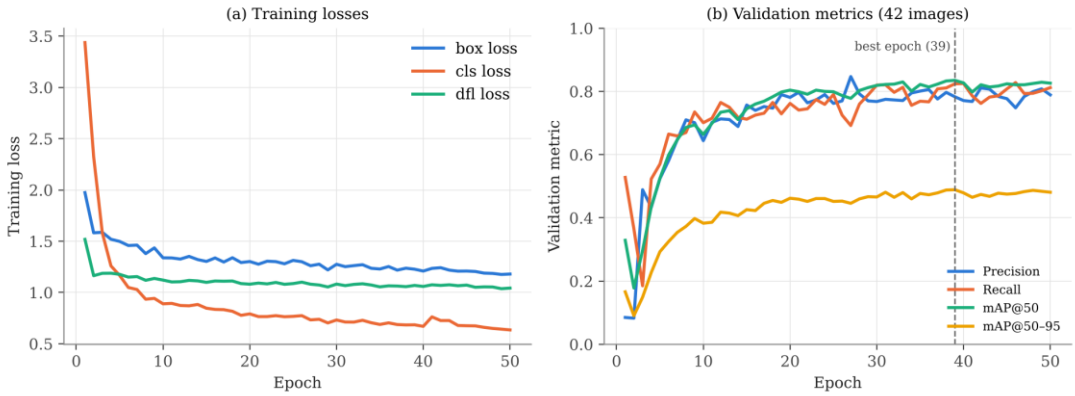


Figure 6.1: YOLO11n training losses and validation metrics per epoch (best checkpoint at epoch 39)

6.1.2 Detector validation performance

On the 42-image validation split the best checkpoint achieved an overall precision of 0.782, recall of 0.824, mAP@50 of 0.835 and mAP@50–95 of 0.489 (Table 6.1, Figure 6.2). Performance differs sharply between classes: Palm reached 0.947 mAP@50 and 0.937 recall, whereas the generic Tree class reached only 0.722 mAP@50 and 0.710 recall.

Table 6.1: YOLO11n validation performance of the best checkpoint

Class	Images	Instances	Precision	Recall	mAP@50	mAP@50–95
Palm	41	1,357	0.916	0.937	0.947	0.615
Tree	31	259	0.649	0.710	0.722	0.364
All	42	1,616	0.782	0.824	0.835	0.489

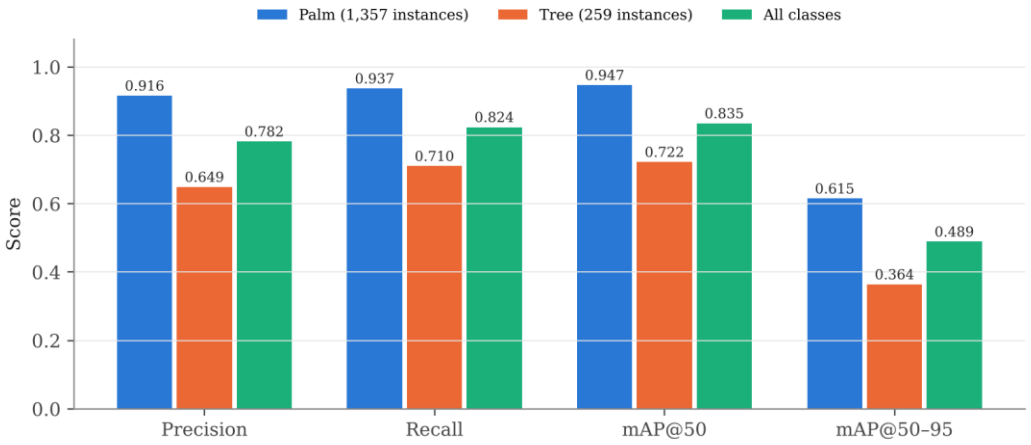


Figure 6.2: Per-class validation metrics of the best YOLO11n checkpoint

The large gap between mAP@50 and mAP@50–95 shows that the detector finds trees reliably but draws boxes less tightly than the strict IoU thresholds demand. This matters downstream because box extent defines where the vegetation mask is searched.

6.2 Product and technical results

6.2.1 Detection on the demonstration image

On the 4864 × 3648-pixel demonstration image the detector returned 86 palm/tree instances with confidences from 0.35 to 0.90 (mean 0.66) (Figure 6.3). The mean canopy radius derived from the boxes was 148.2 px.

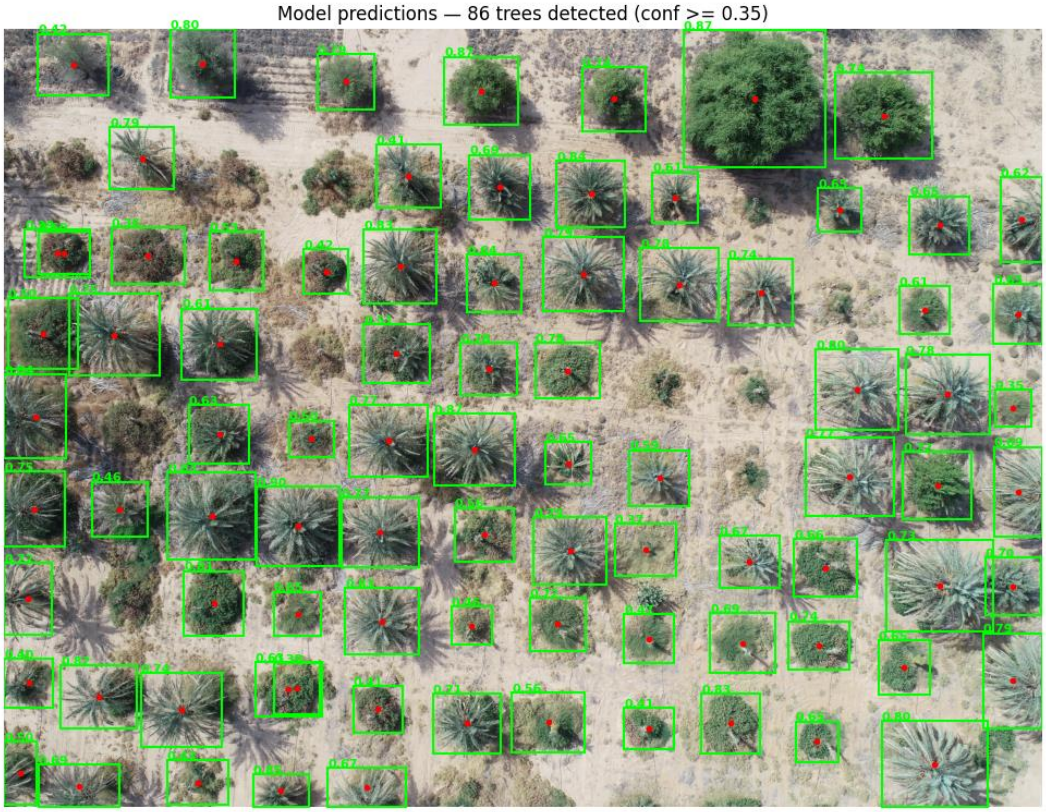


Figure 6.3: YOLO11n detections on the demonstration image (86 instances, confidence ≥ 0.35)

6.2.2 Baseline and refined utilization

Circular buffers with the 148.2-pixel mean canopy-radius proxy produced a baseline utilization estimate of 32.29% and an available-area estimate of 67.71% (Figure 6.4). Applying the ExG-based vegetation mask reduced estimated utilization to 23.81% and increased raw available image area to 76.19% (Figure 6.5), a difference of 8.48 percentage points. The sanity-check visualization shows that the refined mask follows visible canopy structure more closely than complete circular buffers in many detected regions; however, this represents a methodological refinement rather than a measured accuracy improvement because no manually annotated canopy segmentation ground truth was available.

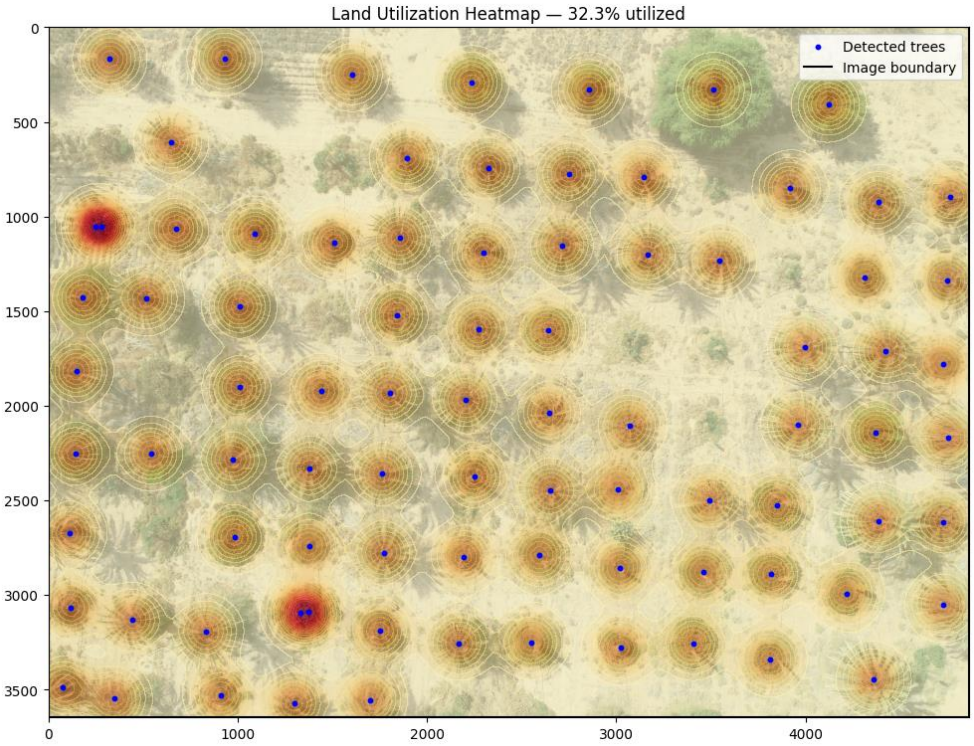


Figure 6.4: Baseline circular-buffer utilization with detection-density heat map (32.3% utilized)

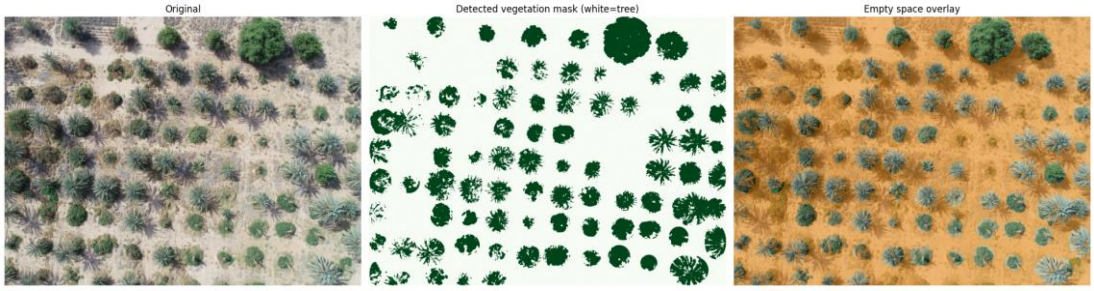


Figure 6.5: Mask sanity check: original image, refined vegetation mask and free-space overlay

6.2.3 Free-space pockets and approximate capacity

The EDT and local-maximum search found 172 candidate pockets with clearance of at least 20 px. The ten most open pockets had clearances between 243.0 and 388.4 px (Figure 6.6; coordinates in Appendix A). Applying the greedy algorithm with S = 148.2 px reduced this to 55 proposed additional positions (Figure 6.7).

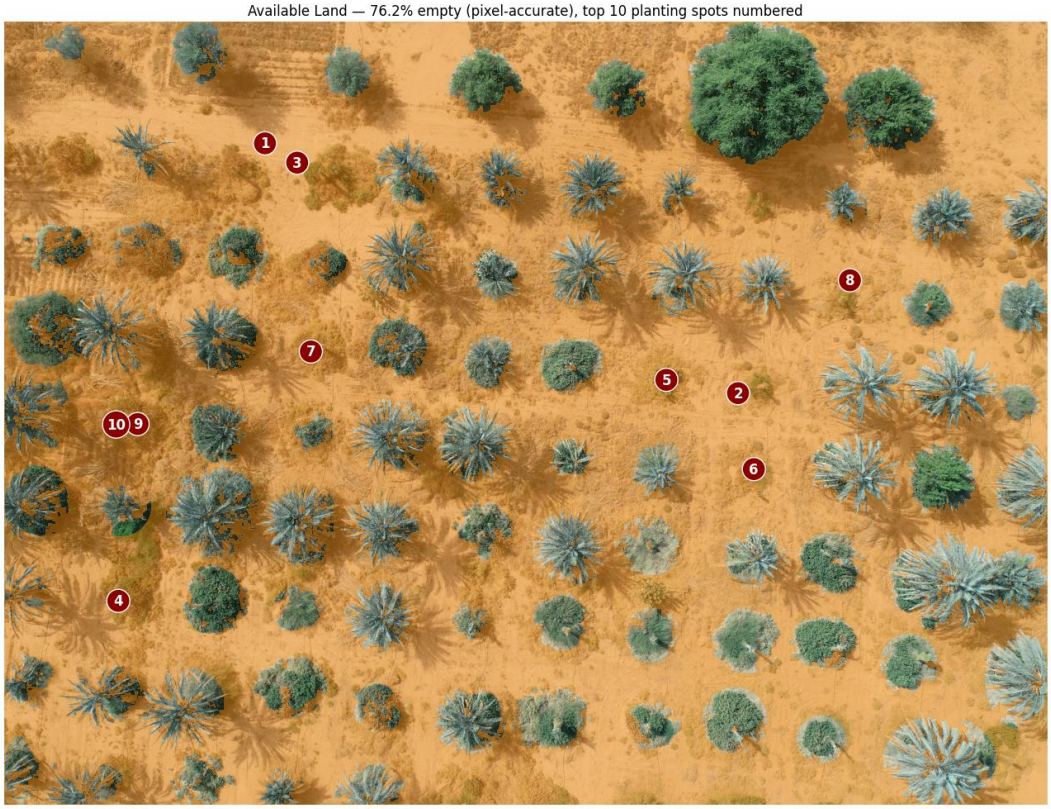


Figure 6.6: Refined free-space map (76.2% available) with the ten highest-clearance pockets numbered

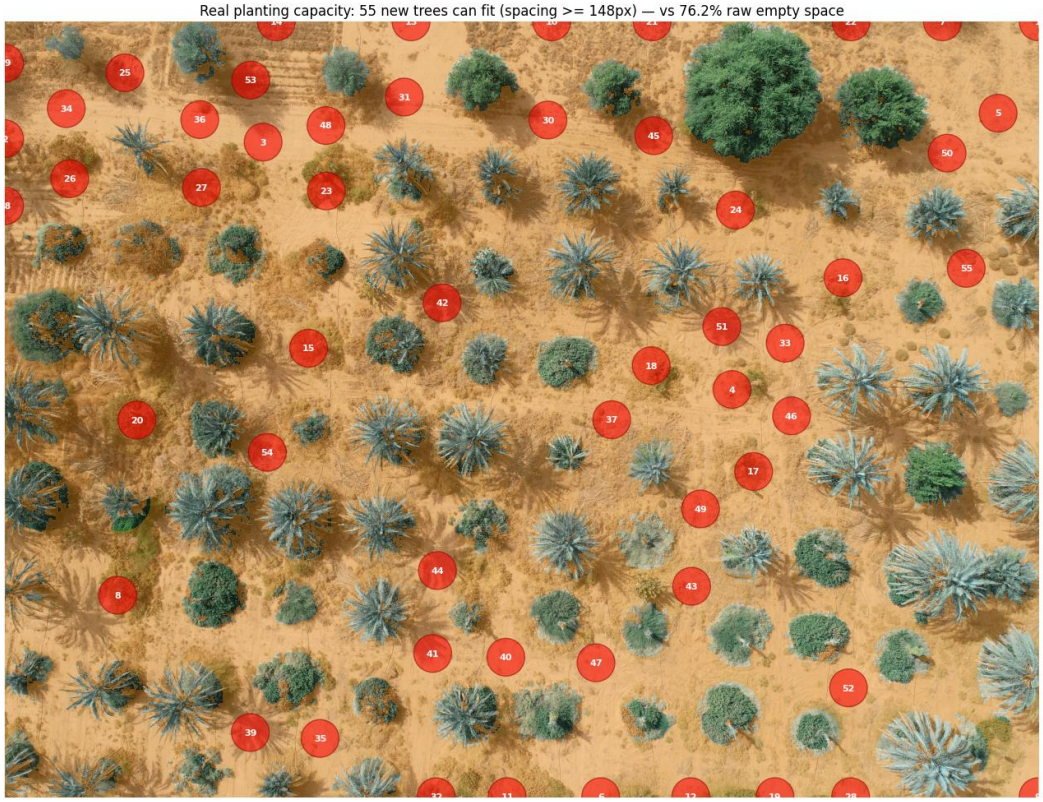


Figure 6.7: Greedy capacity estimate: 55 proposed positions at a 148.2-px spacing proxy

Table 6.2 and Figure 6.8 summarise the land-analysis outputs.

Table 6.2: Land-analysis results on the demonstration image

Output	Value
Image size	4864 × 3648 px
Detected Palm/Tree instances	86
Confidence range / mean	0.35–0.90 / 0.66
Mean canopy radius (spacing proxy S)	148.2 px
Baseline utilized / available	32.29% / 67.71%
Refined utilized / available	23.81% / 76.19%
Candidate free-space pockets	172
Top-10 pocket clearance	243.0–388.4 px
Approximate additional positions (greedy)	55

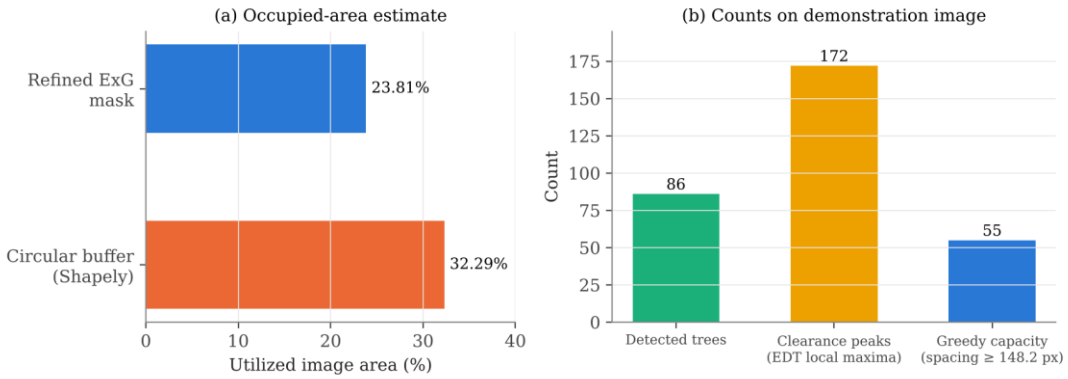


Figure 6.8: Summary of land-analysis outputs: (a) occupied-area estimates; (b) detections, clearance peaks and greedy capacity

The progression from 76.19% available area, to 172 pockets, to 55 positions is the central product result: most free pixels lie in gaps too narrow for a new tree once spacing from canopies and between new trees is enforced.

6.3 User or stakeholder validation

No formal user or stakeholder validation was carried out within the notebook. There is no recorded review by Coconut Research Institute staff, no usability test with plantation managers and no field check of the 55 proposed positions. The results are therefore technical and proof-of-method only. Structured validation with these stakeholders, including a review of the outputs by domain experts and a field check of a sample of proposed positions, is therefore identified as future work (Section 9.4).

6.4 Integration results

The pipeline produced all the artefacts needed for integration: a compact trained model, annotated detection and free-space images, and CSV files of candidate pockets and proposed positions. End-to-end metrics such as API latency, upload limits and concurrent-user throughput were not measured in the notebook and remain product-level validation tasks. No measurements from a deployed platform were available at the time of writing, so integration performance is not reported here and is left to the group-level system evaluation.

6.5 Summary of findings

The findings answer the four research questions at two different levels of evidence. At detector level, YOLO11n achieved an overall mAP@50 of 0.835 and mAP@50–95 of 0.489, with substantially stronger performance for Palm (mAP@50 0.947; recall 0.937) than for the more heterogeneous Tree class (mAP@50 0.722; recall 0.710). These results establish measurable object-localisation performance on the 42-image validation set, while also showing that class-wise performance should not be hidden behind a single aggregate value.

At pipeline level, the demonstration image illustrates how detections can be transformed into progressively more constrained land-use information. The circular baseline estimated 32.29% utilization, whereas the refined ExG mask estimated 23.81% utilization and 76.19% raw available area. Distance-based analysis identified 172 candidate free-space pockets, and the greedy algorithm reduced these to approximately 55 spacing-aware proposed positions under the 148.2-pixel proxy. The result demonstrates that raw available area and candidate-peak count are not equivalent to planting capacity. Because the analysis remains in pixel space and was demonstrated on one image, these values are proof-of-method outputs rather than field-generalized capacity estimates.

7 DISCUSSION

7.1 Interpretation of findings

Detector. An mAP@50 of 0.835 indicates useful detection at moderate overlap, while 0.489 mAP@50–95 shows that tight localisation is much harder. This pattern fits aerial canopies, whose fronds have diffuse, overlapping edges and widely varying size. Localisation error matters here because the box defines the region in which ExG is evaluated; the 15-px padding reduces the risk of cutting off fronds, and the ellipse limits the pixels picked up in the corners.

The per-class results reveal that the overall score hides a substantial difference. Palm, the majority class with 1,357 validation instances, is detected with 0.937 recall, whereas Tree, with only 259 instances and much more varied appearance, reaches 0.710 recall. For a coconut-focused deployment the relevant question is whether a locally trained coconut class can be detected consistently across palm age, cultivar, season and imaging conditions — something this dataset cannot answer.

Utilization. The change from 32.29% circular utilization to 23.81% refined utilization should be interpreted as a change in estimation method, not as a proven increase in accuracy. Complete circular buffers visibly include soil and gaps between fronds, whereas the ExG-based mask retains only pixels satisfying the vegetation rule within padded, ellipse-constrained detection regions. At the same time, some pale or dry foliage may not satisfy ExG > 15, and no ground-truth canopy masks were available to quantify false inclusion or omission. The refined mask was therefore adopted because it is more spatially selective for downstream clearance analysis, while its absolute utilization accuracy remains unverified.

Capacity. The gap between 76.19% free area and 55 positions is the main insight of the work: area is not capacity. Free pixels are often narrow or close to canopies, and the distance-transform and greedy stages add the spatial structure that a percentage lacks. The count is, however, sensitive to the spacing value: because new positions must be at least S from canopy and 2S from each other, doubling S would reduce the count substantially. Visual inspection of Figure 6.7 also shows several positions

placed on the top and bottom image edges, which the algorithm treats as open ground.

7.2 Comparison with literature baselines and existing solutions

Ammar et al. reported strong detection results with several deep detectors on the source imagery [4]. The YOLO11n figures here should not be ranked against theirs, because the model, data split, training settings and evaluation protocol differ. Their study instead confirms that object detection is a sound foundation for aerial palm analysis; this thesis extends that foundation towards occupancy and free-space estimation, which they did not address.

Using ExG follows established vegetation-index practice [8], [9]. Compared with a learned segmentation model it requires no pixel labels and is fully interpretable, but it is more sensitive to foliage colour and lighting, as observed with the pale fronds. The EDT is a standard operation [11]; its contribution here is to turn the mask into a clearance surface from which pockets can be ranked, while the greedy stage adds a spacing constraint that a peak list alone does not enforce. Compared with manual interpretation, the pipeline produces the same result every time for the same image and settings and leaves an audit trail of intermediate outputs.

7.3 Research contribution

The main research contribution is an evaluated hybrid pipeline that links deep object detection to interpretable, image-space land-utilization analysis, with evidence for each of three stages: detector performance, vegetation-mask refinement of geometric occupancy, and spacing-constrained free-space and capacity estimation. The contribution lies in the integration and evaluation of these methods rather than in new algorithms.

A second contribution is the explicit treatment of uncertainty: validation metrics are kept separate from single-image outputs, raw available area from plantable capacity, and pixel distance from real-world spacing. The analysis also identifies concrete, testable sources of bias in such pipelines — image-edge effects, missed detections counted as free space, and colour-threshold under-segmentation — which future work can measure and correct.

7.4 Product contribution

At product level the component produces outputs that non-specialists can understand: a picture of detected trees, a utilized/available percentage, a highlighted free-space map, ranked pocket coordinates and an approximate count of additional positions. CSV export supports further processing and reporting, and the 5.2 MB model is light enough for a modest server. The modular design gives a clear upgrade path to local data, learned segmentation and metric calibration.

7.5 Limitations and threats to validity

Data and generalisation.

1. The detector was trained on a public palm/tree dataset from outside Sri Lanka, not on coconut gene-bank imagery.
2. The classes are Palm and Tree; detections cannot be assumed to be coconut palms.
3. Only 279 annotated images were used, with 42 for validation; there was no separate annotated test set for the detector.
4. The Tree class is under-represented and performs notably worse than Palm.

Measurement.

1. The ExG threshold (15) was set empirically; pale, dry or blue-grey foliage may fall below it, and lighting changes will shift it.
2. No canopy ground-truth masks were available, so mask accuracy and the true utilization are unknown.
3. The ellipse constraint approximates canopy shape and can clip or include pixels for irregular trees.
4. Any tree missed by the detector is absent from the canopy mask and is therefore treated as free image area. In addition, vegetation outside detected Palm/Tree regions is intentionally excluded from occupancy calculations. Both effects can overestimate available area if the omitted vegetation represents a real planting constraint.

5. The image border is not treated as an obstacle, so positions may be proposed at the very edge of the image.

Capacity model and evaluation scope.

1. All calculations are in pixels; no GSD or metric scale was available.
2. The 148.2-px spacing is a proxy derived from the mean canopy radius, not an agronomic spacing standard.
3. Greedy placement is a heuristic and does not guarantee the maximum number of positions.
4. Soil, slope, drainage, irrigation, roads, plot boundaries, root zones and accession-specific spacing are not modelled.
5. The full land analysis was demonstrated on one image, so robustness across images, sites and seasons was not measured.
6. No stakeholder, expert or field validation was recorded.

7.6 Implications

The component shows a feasible path for turning routine aerial images into more useful plantation-management information without a separate trained model for every stage. Once calibrated to local imagery with known GSD and verified spacing, the approach could help researchers and managers screen large areas quickly, flag under-used zones, prioritise field visits and compare utilization between survey dates. Its proper role is decision support: it narrows down where experts should look, while planting decisions remain with field measurement and agronomic judgement.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The value proposition is fast, repeatable screening of aerial plantation imagery to support land-use decisions. Potential users include coconut research stations and gene-bank curators, estate and smallholder plantation managers, agricultural extension services, and drone-survey providers that already capture imagery. The system would give a first-pass estimate of occupied area and usable gaps before a field team measures in detail.

This value proposition is conceptual: willingness to pay, adoption barriers, user requirements and field-validation costs were not studied, so any commercial positioning should follow user interviews and pilot deployments.

8.2 Market and competitive position

The component competes indirectly with manual field surveys, visual interpretation of aerial imagery, generic object-detection services, and specialised agricultural image-analysis or surveying solutions. Its distinguishing feature is the combination of tree detection, lightweight vegetation masking, clearance analysis, and approximate spacing-aware capacity estimation while preserving inspectable intermediate outputs. However, the current prototype lacks real-world scale calibration and local coconut-specific validation, so it should not yet be positioned as a substitute for professional field surveying or expert plantation planning.

8.3 Business or sustainability model

Two models appear feasible: institutional deployment as part of a research-management platform maintained by a coconut research body, or a service in which users upload aerial images and receive reports. Costs would include local data collection and annotation, model retraining, hosting, maintenance and expert validation. Because the detector is small, inference infrastructure could remain modest, but processing of large high-resolution mosaics would need performance testing before costs can be estimated.

8.4 Scalability intellectual property and compliance

Scaling up requires handling large orthomosaics by tiling, supporting different resolutions, and avoiding double-counting across overlapping frames. The dataset and the Ultralytics software are distributed under their own licences, and commercial use must check licence compatibility and attribution obligations. Ownership of the code, trained weights and integration logic among the student team and the institution should be documented. Collecting local aerial data may raise operational, privacy and aviation-regulation requirements depending on site and capture method.

8.5 Contribution to the Sustainable Development Goals

The work has potential links to the United Nations Sustainable Development Goals [21]: SDG 2 (Zero Hunger) through support for agricultural productivity and genetic-resource management; SDG 9 (Industry, Innovation and Infrastructure) through digital innovation in agriculture; SDG 12 (Responsible Consumption and Production) through more efficient use of plantation land; and SDG 15 (Life on Land) through support for conserving plant genetic resources. These are potential pathways, not measured impacts.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis developed an aerial-image component that combines YOLO11n object detection with classical image processing and spatial analysis to estimate land utilization and approximate planting capacity. Trained on 279 aerial images with 10,398 Palm/Tree boxes, the detector reached 0.835 mAP@50, 0.489 mAP@50–95, 0.782 precision and 0.824 recall on the validation split, with strong palm detection (0.947 mAP@50).

The work then moved beyond counting. A refined ExG vegetation mask lowered the estimated utilization of the demonstration image from 32.29% (circular buffers) to 23.81%; a Euclidean distance transform found 172 free-space pockets; and greedy, spacing-aware placement estimated about 55 additional positions. The results show that it is technically feasible to turn aerial detections into interpretable utilization and capacity outputs, and equally that raw free area greatly overstates real capacity. Because all estimates are in pixel space, they must be calibrated and validated in the field before they are used to plan planting.

9.2 Achievement of objectives

Table 9.1 maps each objective from Section 1.5 to its status and the evidence reported in this thesis.

Table 9.1: Achievement of the individual research objectives

Objective	Status	Evidence
1. Prepare YOLO dataset	Achieved	279 images, 10,398 boxes; 237/42 image-level split; 0 skipped
2. Train and evaluate detector	Achieved	YOLO11n, 50 epochs; P 0.782, R 0.824, mAP@50 0.835, mAP@50–95 0.489
3. Baseline occupancy	Achieved	Shapely buffers; 32.29% utilized
4. Refined occupancy	Achieved (not ground-truth validated)	ExG + ellipse + morphology; 23.81% utilized
5. Free-space pockets	Achieved	EDT + peak_local_max; 172 pockets
6. Approximate capacity	Achieved with limitations	Greedy placement; 55 positions at 148.2-px proxy
7. Export for integration	Achieved at artefact level	palm_detector_best.pt, CSV and PNG outputs

9.3 Individual research and product contributions

The individual contribution spans research design, deep-learning experimentation, image-processing development, spatial-analysis logic, evaluation, visualisation and preparation of deployable artefacts. It produced both a trained detector and a transparent sequence of post-processing stages, and its most important outcome is the step-by-step conversion of object detections into increasingly useful land-management information while keeping the evidence for each step open to inspection.

9.4 Future directions

1. Collect coconut aerial imagery from several Sri Lankan sites and annotate a dedicated coconut class.
2. Create canopy ground-truth masks and compare ExG with semantic or instance segmentation models.
3. Use imagery with known GSD (or ground control points) so that distances and areas are reported in metres and square metres, and capacity in palms per hectare.

4. Replace the canopy-radius proxy with agronomically verified coconut spacing rules and layouts (square or triangular).
5. Add plot-boundary masks and exclude roads, buildings, waterways and other non-plantable surfaces; treat the image edge as a boundary.
6. Add terrain, soil, drainage, irrigation and accession-specific constraints where data exist.
7. Run a controlled comparison of YOLO11 model sizes and alternative architectures, with a separate annotated test set.
8. Test the full land-analysis pipeline on a representative set of independent images and report variability.
9. Compare greedy placement with optimisation or grid-packing methods that may achieve higher capacity.
10. Carry out expert review and field checks of proposed positions.
11. Measure inference time, memory use and end-to-end response time in the deployed web service.
12. Support repeated surveys so utilization changes can be tracked over time.

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GLOSSARY

Bounding box – An axis-aligned rectangle (x1, y1, x2, y2) that localises one detected object in an image.

Canopy – The crown of a tree as seen from above; for palms, the star-shaped spread of fronds.

Clearance – For a free pixel, its Euclidean distance (in pixels) to the nearest blocked pixel.

Confidence threshold – The minimum detector score (0.35 in this work) for a prediction to be kept.

Early stopping (patience) – Ending training when the validation fitness has not improved for a set number of epochs.

Fine-tuning / transfer learning – Continuing the training of a model pretrained on a large dataset using a smaller task-specific dataset.

Greedy placement – An iterative heuristic that always takes the currently best option (here, the most open spot) without revisiting earlier choices.

Ground sampling distance – The ground distance represented by one image pixel (e.g. centimetres per pixel).

Morphological closing / opening – Binary image operations that respectively fill small holes and remove small isolated specks.

Vegetation mask – A binary image in which each pixel is labelled vegetation (occupied) or not (available).

APPENDICES

Appendix A Extended results and configuration

A.1 Implementation constants

Table A.1 lists the configuration constants used by the land-analysis cells.

Table A.1: Key implementation constants used in the notebook

Constant	Value
MODEL_PATH	/kaggle/working/runs/palm_detector/weights/best.pt
TEST_IMAGE_PATH	.../Palm-Counting-349images/test/ck2euxc9kxgvm07486g2d5pid.jpg
CONFIDENCE_THRESHOLD	0.35
BOX_PADDING_PX	15
EXCESS_GREEN_THRESHOLD	15
USE_ELLIPSE_CONSTRAINT / ELLIPSE_SCALE	True / 1.0
GRID_DOWNSAMPLE	2
MIN_DISTANCE_BETWEEN_POCKETS	15 (grid cells)
MIN_CLEARANCE_PX	20
TOP_N_POCKETS	10
REQUIRED_SPACING_PX	None → auto (148.2 px)
MAX_NEW_TREES_TO_FIND	500

A.2 Top ten free-space pockets on the demonstration image

Table A.2: Ten highest-clearance candidate pockets (full-resolution pixel coordinates)

Rank	x (px)	y (px)	Clearance (px)
1	1216	566	388.40
2	3422	1730	377.92
3	1366	656	365.02
4	532	2698	300.24
5	3088	1668	294.00
6	3496	2084	272.26
7	1430	1536	262.49
8	3944	1204	254.50
9	622	1874	244.75
10	522	1876	243.02

A.3 Selected per-epoch validation metrics

Table A.3: Validation metrics at selected epochs (from the training log)

Epoch	Precision	Recall	mAP@50	mAP@50–95
1	0.085	0.528	0.329	0.166
5	0.524	0.570	0.525	0.293
10	0.644	0.701	0.664	0.383
15	0.757	0.712	0.745	0.426
20	0.781	0.762	0.804	0.462
25	0.762	0.790	0.799	0.452
30	0.768	0.820	0.819	0.466
35	0.801	0.769	0.822	0.477
39 (best)	0.783	0.824	0.835	0.489
40	0.771	0.825	0.827	0.479
45	0.777	0.808	0.824	0.475
50	0.789	0.812	0.826	0.481

Note: the validation printed at the end of model.train() reported overall P = 0.781 and R = 0.830 for the same best checkpoint, while the separate model.val() call reported

P = 0.782 and R = 0.824; mAP values were identical (0.835 / 0.489). The small precision/recall difference comes from the validator's confidence-threshold selection between runs. The model.val() values are used throughout this thesis.

The full notebook (coconut-land-detection.ipynb), the trained weights (palm_detector_best.pt) and the exported CSV files (planting_candidate_spots.csv and real_planting_capacity.csv) are available from the candidate on request.

Appendix B Academic integrity and AI use disclosure

Generative AI tools were used as drafting and language-support aids during preparation and consolidation of this thesis. This included ChatGPT (OpenAI) and, as recorded in an earlier thesis draft, Claude (Anthropic). Their use was limited to activities such as organising chapter structure, improving academic wording, comparing draft versions, and checking narrative consistency against the supplied implementation evidence. All technical values, including dataset counts, training parameters, software versions, validation metrics, image-analysis constants, and reported land-utilization outputs, were verified against the coconut-land-detection.ipynb notebook or the two source thesis drafts before inclusion. Generative AI was not used to generate the experimental dataset, train the YOLO11n model, execute validation, or create the reported numerical results. The student remains responsible for final verification of originality, citation accuracy, technical correctness, and compliance with the current module policy.

The exact disclosure format, representative prompts, and tool-identification requirements should follow the current IT4010/module instructions at the time of submission. Where institutional policy requires a prompt log or a list of affected sections, those records should be attached using verified information from the student's actual workflow.